

»» Chapter XVIII ««

RIEMANN SURFACES

PREFATORY REMARKS

We terminate this book with a study of the concept of a Riemann surface and its relation to classical complex function theory. The modern theory of Riemann surfaces initiated by Hermann Weyl in the first edition of his *Idee der Riemannsche Fläche* [128] is actively cultivated at the present time. The collection of monographs on the subject is extensive and the interested reader will not want for material to study. To state in very rough terms the role played by a Riemann surface, it is that of a natural kind of domain for analytic, meromorphic, and harmonic functions. The reader must be conscious of the limitations of the classical theory based on plane domains. It suffices to recall that a function analytic on an annulus which does not take the value zero need not have an analytic logarithm, though there is no difficulty locally. It was Riemann's profound insight to observe that there were a natural cohesiveness and an organic unity in the theory of analytic functions to which adhering to the plane did violence. The introduction of the concept of a Riemann surface to assure the theory of analytic functions its inherent harmony is one of his many illustrious achievements.

It is not our intention to present here a condensed account of the modern theory of Riemann surfaces. For a suitable comprehensive treatment of the subject the preliminaries are extensive. Our goal is more specific. It is to show how the theory of analytic continuation appears as a bridge between classical complex function theory and modern Riemann surface theory. A striking feature of the bridging relation is the remarkable result, due to Riemann and

Klein in the compact case and to Koebe in the noncompact case, that the conformal equivalence classes (we refer to the “natural” equivalence relation between Riemann surfaces) contain exemplars drawn from the theory of analytic continuation as refined in the present chapter. We shall treat the compact case here with the aid of the Perron method. For the noncompact case a proof along comparable lines is available. However it will be omitted because it is somewhat lengthy (cf. [114]).

The study of compact Riemann surfaces is of particular interest since it appears as a natural framework for the study of algebraic equations of the form

$$[w(z)]^{n+1} + \sum_0^n a_k(z) [w(z)]^k = 0,$$

where the a_k are rational functions. By the theory of analytic continuation we shall achieve a synthesis and fusion of the local solutions leading inevitably, one might say, to compact Riemann surfaces.

The chapter concludes on an algebraic note with a theorem (Theorem 7.1) leading at once to the conclusion that conformal equivalence of compact Riemann surfaces is mirrored in the algebraic equivalence of their fields of meromorphic functions relative to constant-preserving isomorphisms.

1. RIEMANN SURFACES IN THE SENSE OF WEYL-RADÓ

A Riemann surface in the sense of Weyl-Radó is an ordered pair (S, Θ) where (1) S is a connected Hausdorff space, (2) Θ is a family of univalent continuous maps of regions of $\mathbb{C}$ into S which are open (that is, map open sets of their domains into open sets of S), the set of images of the members of Θ having union S , (3) Φ , the set of the inverses of the members of Θ , satisfies the condition: when $\varphi, \psi \in \Phi$ have domains with a nonempty intersection, say D , then

$$\{(\varphi(p), \psi(p)): p \in D\}$$

is analytic on $\varphi(D)$. This is what is meant by the notion of an abstract Riemann surface as it has developed from the concept originally introduced by Weyl (which involved triangulability features turning out subsequently to be dispensable thanks to a result of T. Radó [147]). The definitions commonly given today differ slightly in formulation from one another but are equivalent to the one given above. It is to be observed that fundamental to the whole idea is the connection property (3). The family Θ is said to be a *conformal structure* on S . A univalent open continuous map α of a region of $\mathbb{C}$ into a

Riemann surface S is termed a *uniformizer* provided that $(S, \Theta \cup \{\alpha\})$ is a Riemann surface. It is to be noted that the established usages of this book concerning underlying sets will prevail.

EXERCISE

1.1. A univalent continuous map α of a region of $\mathbb{C}$ into a Riemann surface S is a uniformizer if and only if for each point a in the domain of α there exist a neighborhood V of a and $\varphi \in \Phi$ whose domain contains $\alpha(V)$ such that $\varphi \circ (\alpha|_V)$ is analytic at a .

We recall that analyticity and harmonicity are preserved under composition with analytic functions. Thanks to this fact the concepts of analytic, harmonic, and meromorphic functions enter naturally into the study of Riemann surfaces. Given $f: V \rightarrow \hat{\mathbb{C}}$ where V is a neighborhood of a point p of a Riemann surface S , we say that f is *meromorphic at p* provided that there exists a uniformizer α mapping its domain into V , $\alpha(a) = p$, such that $f \circ \alpha$ is meromorphic at a in the sense of §4, Ch. VI. When $f(p) \neq \infty$ and f is meromorphic at p , we say that f is *analytic at p* . The notion of harmonicity at a point is analogously introduced in terms of the existence of a uniformizer leading by composition to a harmonic function in the sense of the complex plane. It is immediate that if $f \circ \alpha$ is meromorphic at a for one such uniformizer, then this property is enjoyed by all such uniformizers, when we note that condition (3) of the definition of a Riemann surface yields the result that if β is also a uniformizer for S , $\beta(b) = p$, then there exists g analytic at b , $g(b) = a$, such that $\beta(z) = \alpha[g(z)]$, z near b . A similar remark holds for harmonicity.

For a function f meromorphic at p the multiplicity of $f \circ \alpha$ at a is independent of allowed α . We denote it by $n(p; f)$. In the same way we see that the divisor of $f \circ \alpha$ at a is independent of allowed α . We denote it by $\partial_f(p)$.

We say that $f: S \rightarrow \hat{\mathbb{C}}$ is *meromorphic (analytic) on S* provided that f is meromorphic (analytic) at each point of S . A similar definition holds for "harmonic on S ."

EXERCISE

1.2. Let A be a region of $\mathbb{C}$ with its usual topology. Let Θ consist of exactly the identity map of A onto itself. Then (A, Θ) is a Riemann surface. Here the definitions of this section are concordant with those given earlier for the classical plane theory.

Let S be $\hat{\mathbb{C}}$ and let Θ consist of z and $1/z$, each with domain $\mathbb{C}$. Then (S, Θ) is a Riemann surface. Further the terms "meromorphic at a point," etc., agree with those used earlier.

Analytic maps. Let there be given $f: S \rightarrow T$ where S and T are Riemann surfaces. We say that f is *analytic at* $p \in S$ provided that there exist uniformizers α for S and β for T such that p is in the image of α , the image of $f \circ \alpha$ is in the image of β , and

$$\beta^{-1} \circ (f \circ \alpha) \quad (1.1)$$

is analytic at $\alpha^{-1}(p)$ in the ordinary sense. We see using the connection property of uniformizers that (1.1) is analytic at $\alpha^{-1}(p)$ for all allowed (α, β) when it is for one ordered pair (α, β) . Since analyticity at p implies continuity for f at p , when f is analytic at p , $\beta^{-1}\{f[\alpha(z)]\}$ is defined in a neighborhood of $\alpha^{-1}(p)$ and is analytic at $\alpha^{-1}(p)$ whenever α is a uniformizer for S with image containing p and β is a uniformizer for T with image containing $f(p)$. It is readily seen that the conclusion of the preceding sentence for some admitted (α, β) implies analyticity of f at p .† We see that the multiplicity at $\alpha^{-1}(p)$ of the function given by (1.1) is independent of allowed (α, β) for f analytic at p . We term the common value the *multiplicity of f at p* and denote it by $n(p; f)$. The map f is said to be *analytic on* S provided that f is analytic at each point of S . We also say in this case that f is an *analytic map of S into (onto) T* . We say that S is *conformally equivalent* to a Riemann surface T provided that there exists a univalent analytic map of S onto T . Referring to the definitions of maps analytic at a point and on a Riemann surface, we see that analyticity at a point is preserved by composition and that the inverse of a univalent analytic map f of S onto T is an analytic map of T onto S . For the latter assertion it is to be noted that f is an open map. We are led to the conclusion that conformal equivalence is a bona fide equivalence relation for Riemann surfaces. It is the primordial classifying notion in Riemann surface theory.

EXERCISES

- 1.3. Determine the effect of composition on the multiplicity of an analytic map.
 1.4. Let f be an analytic map of a Riemann surface S into a Riemann surface T . Show that either f is constant or else for each $q \in T$ the set $f^{-1}(\{q\})$ clusters at no point of S .

† It is to be noted that a function that is meromorphic at a point may qualify for the designation “analytic at the point” in its capacity as a map with image in $\hat{\mathbb{C}}$ taken as a Riemann surface with the conformal structure specified in Ex. 1.2. For maps with images specified to be in $\hat{\mathbb{C}}$ we shall always use the terms “meromorphic” and “analytic” in the senses laid down earlier. Implicit in the use of the term “analytic” in the sense of this paragraph is the presence of the accompanying Riemann surfaces S and T .

2. WEIERSTRASS CLASSES, ANALYTIC ENTITIES

One of our principal objects in this chapter is to examine the theory of analytic continuation in terms of Riemann surface theory. We return to the concepts of an element and of a Weierstrass class introduced in §6, Ch. XII. We consider $\mathfrak{E}$ the set of all elements and proceed as follows: (1) We associate with each element $e = (a, P)$ the map θ_e from the domain of P into $\mathfrak{E}$ which assigns to z in the domain of P the direct continuation with center z of e and note that θ_e is univalent. (2) We introduce a topology on $\mathfrak{E}$ rendering each θ_e an open continuous map of the domain of θ_e into $\mathfrak{E}$ and thereupon verify that the components of $\mathfrak{E}$ are precisely the Weierstrass classes, that each Weierstrass class W together with $\{\theta_e: e \in W\}$ is a Riemann surface, and that the center and value functions are analytic on W and have the property that $c(e), v \circ \theta_e = e, e \in W$.

Development of (2). We must accept among the open sets of $\mathfrak{E}$ the sets $\theta_e(O), e \in \mathfrak{E}, O$ an open subset of the domain of θ_e . Let T denote the smallest topology on $\mathfrak{E}$ containing all such sets. We show: (i) T is a Hausdorff topology; (ii) each θ_e is an open continuous map of its domain into W ; (iii) T is the only topology on W enjoying the property (ii).

(i) Let there be given elements (a, P) and (b, Q) which are distinct. If $a \neq b$, then the images of disjoint small disks centered at a and b with respect to the associated θ maps are disjoint since the first components of the points of the first image set are different from the first components of the second image set. If $a = b$, then $P \neq Q$. Hence the direct continuations of (a, P) and (b, Q) with center z near a are distinct. Hence the images of a small disk centered at a with respect to the two θ maps are disjoint. The Hausdorff property is satisfied by T .

(ii) Given elements e_1 and e_2 and O_k open in the domain of $\theta_{e_k}, k = 1, 2$, we show that $\omega = \theta_{e_1}(O_1) \cap \theta_{e_2}(O_2)$ is a set $\theta_{e_1}(O)$. By induction the same holds true for a finite intersection of this kind, $\bigcap_{1 \leq k \leq n+1} \theta_{e_k}(O_k), n \in \mathbb{N}$. It follows that T consists precisely of the unions of the sets of the form $\theta_e(O)$ specified above. Suppose that $f \in \omega$. Then f is a direct continuation of both e_1 and e_2 . Hence the direct continuations of f with centers sufficiently near $c(f)$ belong to ω . Consequently $\theta_{e_1}^{-1}(\omega)$ is a neighborhood of $c(f)$ and we conclude that $\theta_{e_1}^{-1}(\omega)$ is open. Our assertion follows.† The continuity of θ_e follows on noting that if $\Omega \in T$, then $\theta_e^{-1}(\Omega)$ is by the property of T just established the union of open subsets of the domain of θ_e and hence open.

(iii) Suppose that T_1 is also a topology on $\mathfrak{E}$ rendering each θ_e an open

† We remark that one may even show that if $\omega \neq \emptyset$, then $\omega = \theta_{e_k}(O_1 \cap O_2), k = 1, 2$.

continuous map of its domain into $\mathfrak{E}$. Let $\Omega \in T_1$. Since

$$\Omega = \bigcup_{e \in \mathfrak{E}} \theta_e[\theta_e^{-1}(\Omega)] \in T,$$

we conclude that $T_1 \subset T$ and thereupon that $T_1 = T$.

Let $e \in \mathfrak{E}$. We show that the component K of $\mathfrak{E}$ containing e is just the Weierstrass class $\mathbf{W}$ containing e . Suppose that π is a continuation of e along a path γ issuing from $c(e)$. We verify that π is a continuous map of its domain into $\mathfrak{E}$. Indeed, if s is in the domain of π , then

$$\pi(t) = \theta_{\pi(s)}[\gamma(t)]$$

for t sufficiently near s .[†] Hence $\mathbf{W}$ is arcwise connected and so $\mathbf{W} \subset K$. Now $\mathbf{W}$ is open, for if $f \in \mathbf{W}$, then the image of θ_f is contained in $\mathbf{W}$. Further if $f \in \mathbf{W}$, then the image of θ_f contains a point of $\mathbf{W}$ and so f is contained in $\mathbf{W}$, that is, $\mathbf{W} = \mathbf{W}$. The arcwise connectibility wanted in each of these situations is a consequence of the obvious validity of this property for a disk in $\mathbf{C}$ and of the fact that $\theta_f \circ \gamma$ is a continuation when γ is a path in the domain of θ_f . We conclude by the connectedness of K and the fact that $\mathbf{W}$ is both open and closed that $\mathbf{W} = K$. Thus the Weierstrass classes are precisely the components of $\mathfrak{E}$.

To show that $\Theta = \{\theta_e: e \in \mathbf{W}\}$ is a conformal structure on $\mathbf{W}$, we note that if φ and ψ are the inverses of maps that are members of Θ , and the intersection D of the domains of φ and ψ is not empty, then $\varphi(e) = \psi(e)$, $e \in D$, so that the connecting map $\{(\varphi(e), \psi(e)): e \in D\}$ is the identity map and so obviously is analytic on its domain. Hence $(\mathbf{W}, \Theta)$ is a Riemann surface. Given $e = (a, P) \in \mathbf{W}$, we have $c \circ \theta_e(z) = z$, $v \circ \theta_e(z) = P(z)$, z in the domain of θ_e . Hence not only do we conclude that c and v are analytic on $\mathbf{W}$, but also that $(c(e), v \circ \theta_e) = e$.

Analytic entity (= *analytische Gebilde in the sense of Weyl*). It will be noted that in the above developments relating analytic continuation and Riemann surface theory, which are otherwise harmonious, there is a possible lack of maximality. Thus far no account is taken of poles or of analyticity at ∞ . Another situation that calls for attention is the following. Let $\mathbf{W}$ denote the Weierstrass class containing $(1, S)$, where S is an analytic square root of the identity map of $\Delta(1; 1)$ onto itself. Then no member of $\mathbf{W}$ has 0 as a center. Yet we sense from the behavior of v and c , $c = v^2$, that we are in the presence of an "isolated" situation calling for further examination when we consider the elements of $\mathbf{W}$ with center near 0. In any case, from the point of view of limiting behavior there is great regularity: when $c(e)$ is small, so is $v(e)$.

Before we proceed further we introduce some illustrative exercises.

[†] It is worth noting that a path π in $\mathfrak{E}$ is a continuation along $c \circ \pi$.

EXERCISES

2.1. Let W denote the Weierstrass class containing $(1, P)$ where P is $1/z$ with domain $\Delta(1; 1)$. Show that c maps W univalently onto $C - \{0\}$ and that, in fact, W consists of the elements (a, P_a) , $a \in C - \{0\}$, where P_a is $1/z$ with domain $\Delta(a; |a|)$.

Here we see that W is defective in comparison with the parent function $1/z$ with domain $\hat{C}$, for W cannot take into account 0 and ∞ as centers.

2.2. Let f and g be nonconstant polynomials. Let $a \in C$ be such that $f'(a) \neq 0$. Let $(f(a), h) \in \mathfrak{E}$ satisfy

$$h[f(z)] = g(z),$$

z near a . Show that the Weierstrass class W containing $(f(a), h)$ consists of the elements of the form $(f(w), P_w)$ where $f'(w) \neq 0$ and $P_w[f(z)] = g(z)$, z near w , together with a possible finite set of elements. Show that the valence of c (defined as in the plane theory) takes the same finite value at each point of the set A consisting of $w \in C$ such that $f'(z) \neq 0$ for each $z \in f^{-1}(\{w\})$, that this value is at most the degree of f , and that elsewhere the valence of c is no greater than it is on A . Show that an element of W with center in A may be continued along an arbitrary path in A .

Consider explicitly the cases where $(f(z), g(z))$ is one of the following: (z^2, z) , (z^2, z^2) , (z^2, z^4) , (z^4, z^2) , $(z(z-1)^2, z^2(z-1))$.

Returning to the general situation show that if the valence of c is constant on C , then it is the constant 1 and $g = Q \circ f$, where Q is a polynomial. Hint: Monodromy theorem. Is there a converse?

We propose to consider the deficiencies of a Weierstrass class from the following point of view. Given a Weierstrass class W we ask under what circumstances there exists (S, f, g, φ) where S is a Riemann surface, f and g are meromorphic on S , and φ is an analytic map of W into S , such that

$$c = f \circ \varphi, \quad v = g \circ \varphi. \quad (2.1)$$

In other words, if we can gain (c, v) for W from (f, g) via φ , what are the natures of S, f, g, φ ?

Let us be specific. We first examine the univalence of φ . Suppose that $\varphi(e_1) = \varphi(e_2) = q$. We see that $c(e_1) = c(e_2)$. Let $e_1 = (a, P_1)$, $e_2 = (a, P_2)$. From (2.1) we see that neither f nor φ is constant, and further, that

$$g \circ (\varphi \circ \theta_{e_k}) = v \circ \theta_{e_k} = P_k = P_k \circ [f \circ (\varphi \circ \theta_{e_k})],$$

$k = 1, 2$. Hence $g(p) = P_k[f(p)]$ for p in some neighborhood of q , and consequently, since f enjoys the open mapping property, $P_1(z) = P_2(z)$, z near a . We conclude that $e_1 = e_2$. Thus φ is univalent. From $c = f \circ \varphi$ we see that the multiplicity of f is 1 at each point of $\varphi(W)$.

We introduce the set E consisting of the points p of S at which at least one of the following conditions is fulfilled: $f(p) = \infty$, $g(p) = \infty$, $n(p; f) > 1$. We shall establish the following equality:

$$\varphi(W) = S - E. \quad (2.2)$$

It is immediate that $\varphi(\mathbf{W}) \subset S - E$. We note that there exists a unique map ψ from $S - E$ into $\mathfrak{E}$ such that $f(q)$ is the center of $\psi(q)$ and $g(p) = P_q[f(p)]$, p near q , where P_q is the second component of $\psi(q)$. It is easily verified that ψ is continuous and so its image is contained in a Weierstrass class. That it is $\mathbf{W}$ itself may be seen as follows. Suppose that $e_0 = (a, P) \in \mathbf{W}$ and let $q = \varphi(e_0)$. Because of the conditions imposed on f, g, φ , we have

$$f(p) = \theta_{e_0}^{-1}[\varphi^{-1}(p)]$$

and

$$g(p) = P\{\theta_{e_0}^{-1}[\varphi^{-1}(p)]\}$$

when p is near q . Hence $\psi(q) = e_0$. Given the arbitrariness of e_0 , we see that $\psi \circ \varphi$ is the identity map of $\mathbf{W}$ onto itself. It follows that $\psi(S - E) = \mathbf{W}$ and that $\psi|_{\varphi(\mathbf{W})}$ is univalent.

To complete the proof we show that

$$\overline{\varphi(\mathbf{W})} \cap (S - E) \subset \varphi(\mathbf{W}). \quad (2.3)$$

It will follow that inclusion may be replaced by equality and hence that $\varphi(\mathbf{W}) = S - E$ since $S - E$ is connected. Indeed, $\varphi(\mathbf{W})$ is open. Further $\varphi \circ \psi(p) = p$, $p \in \varphi(\mathbf{W})$, as is easily verified. Hence by the continuity of $\varphi \circ \psi$ we have $\varphi \circ \psi(p) = p$, $p \in \overline{\varphi(\mathbf{W})} \cap (S - E)$. The inclusion (2.3) follows.

We now proceed to show that we may choose (S, f, g, φ) in a maximal fashion relative to $\mathbf{W}$. More exactly, we have the following theorem.

Theorem 2.1: *There exists (S, f, g, φ) standing in the stated relation to a given Weierstrass class $\mathbf{W}$ such that if $(S_1, f_1, g_1, \varphi_1)$ also stands in the same relation to $\mathbf{W}$, there exists a univalent map ψ of S_1 into S , analytic on S_1 , which satisfies*

$$f_1 = f \circ \psi, \quad g_1 = g \circ \psi, \quad \varphi = \psi \circ \varphi_1. \quad (2.4)$$

Such an ordered quadruple (S, f, g, φ) is determined up to a conformal equivalence in the sense that if $(S_2, f_2, g_2, \varphi_2)$ is a second such maximal ordered quadruple, then there exists ψ satisfying the stated conditions with $(S_2, f_2, g_2, \varphi_2)$ replacing (S, f, g, φ) and (S, f, g, φ) replacing $(S_1, f_1, g_1, \varphi_1)$ which maps S onto S_2 .

The proof of this theorem will appear as the end product of the developments that follow. We start with the formulation of the notion of an analytic entity. We consider ordered pairs (f, g) where f and g both have the same domain, a region in $\mathbf{C}$ containing 0, on which each is meromorphic, and where in addition the map assigning $(f(z), g(z))$ to the point z of their common domain is univalent. Such (f, g) will be termed *allowed*. We propose an

equivalence relation between allowed (f, g) by the following stipulation: (f_1, g_1) is said to be *equivalent* to (f, g) provided that there exists φ analytic at 0, $\varphi(0) = 0$, such that

$$f_1(z) = f[\varphi(z)], \quad g_1(z) = g[\varphi(z)],$$

z small. Under these conditions we see that $n(0; \varphi) = 1$ and further that φ is determined locally. That we have to do with a bona fide equivalence relation is a matter of routine verification. We denote the equivalence class containing (f, g) by $e(f, g)$ and the set of such $e(f, g)$, taking into account all allowed (f, g) by $\mathbf{E}$. The members of $\mathbf{E}$ will be termed *meromorphic elements*. We note that $(f(0), g(0))$ is independent of (f, g) in a given equivalence class e . We denote the common value, $f(0)$, of the first components by $C(e)$ and term it the *center of e* ; we denote the common value, $g(0)$, of the second components by $V(e)$ and term it the *value of e* . No confusion should arise from the use of the same terminology for the purposes of Weierstrass classes and analytic entities since in every case it will be clear in what sense the terms are to be taken.

The first part of the program to be carried out parallels that for Weierstrass classes and treats (1) a topology for $\mathbf{E}$, (2) conformal structure for the components of $\mathbf{E}$.

EXERCISE

2.3. If the component f of an allowed (f, g) is not constant, then (f, g) is equivalent to an allowed (f_1, g_1) where $f_1(z) = f(0) + z^{n(0; f)}$, when $f(0) \neq \infty$, and $f(z) = 1/z^{n(0; f)}$ when $f(0) = \infty$. Show that the 0-Laurent coefficients of g_1 are determined up to an $n(0; f)$ th root of unity in the sense that (f_1, g_1) is equivalent to (f_1, h) if and only if there exists an $n(0; f)$ th root of unity, say ω , such that $c_k = \omega^k b_k$ where (b_k) is the 0-Laurent sequence of g_1 and (c_k) is the 0-Laurent sequence of h . Show further that an ordered pair of the form (f_1, G) , where G is meromorphic on the domain of f_1 , is such that the ordered pair

$$(f_1 | \Delta(0; r), G | \Delta(0; r))$$

is allowed for small r if and only if the only positive integer dividing $n(0; f)$ and $k \in \mathbf{Z}$ for which $d_k \neq 0$ is 1, (d_k) being the 0-Laurent sequence of G .

If (f, g) is allowed and f is constant, then (f, g) is equivalent to (f, g_1) where $g_1(z) = g(0) + z$ when $g(0) \neq \infty$ and $g_1(z) = 1/z$ when $g(0) = \infty$.

The topology on $\mathbf{E}$. Given an allowed pair (f, g) we introduce a map $\theta_{(f, g)}$ of the common domain Ω of f and g into $\mathbf{E}$ as follows. Given $a \in \Omega$, let $\tau_a = \sigma_a^{-1}$ where σ_a is the function $z - a$ with domain Ω . We define $\theta_{(f, g)}(a)$ as $e(f \circ \tau_a, g \circ \tau_a)$. We call the $\theta_{(f, g)}$ so introduced θ maps. To anticipate, we remark that the $\theta_{(f, g)}$ will serve to furnish the conformal structures on the components of $\mathbf{E}$. We take as the topology T the topology generated by the

images with respect to the θ maps of the open subsets of their domains. The following lemma serves to show that T is a Hausdorff topology.

Lemma 2.1: *Let m and n be positive integers and let f and g be analytic on $\Delta(0; r)$ such that the maps $(z^m, f(z))$ and $(z^n, g(z))$ with domain $\Delta(0; r)$ are univalent. Suppose that sequences $(z_k)_{k \in \mathbb{N}}$, $(w_k)_{k \in \mathbb{N}}$ exist which satisfy: $0 < |z_k|$, $|w_k| < r$, $z_k^m = w_k^n$, $f(z_k) = g(w_k)$, $k \in \mathbb{N}$, as well as $\lim_{k \rightarrow \infty} z_k = 0$. Then $m = n$ and there exists an m th root of unity, say ω , such that $f(z) = g(\omega z)$, $|z| < r$. In particular, (z^m, f) and (z^n, g) are equivalent.*

PROOF: Let t_k be an n th root of z_k . Then $w_k t_k^{-m}$ is an n th root of unity which we assume (as we may by the use of subsequences when necessary) to be the same, say ω , for all k . We have $f(t_k^n) = g(\omega t_k^m)$, $k \in \mathbb{N}$, and hence

$$((t^n)^m, f(t^n)) = ((\omega t^m)^n, g(\omega t^m)), \quad (2.5)$$

$|t| < \rho$ where $\rho = +\infty$ when $r = +\infty$ and otherwise is $\min\{r^{1/m}, r^{1/n}\}$. Since the function $t \rightarrow ((t^n)^m, f(t^n))$, $0 < |t| < \rho$, takes each of its image points at exactly n distinct points and a corresponding statement holds for $t \rightarrow ((\omega t^m)^n, g(\omega t^m))$, $0 < |t| < \rho$, with m replacing n , we see that $m = n$ and thereupon from (2.5) that

$$(z^m, f(z)) = ((\omega z)^m, g(\omega z)), \quad |z| < r.$$

We conclude the assertions of the lemma.

The topology T satisfies the Hausdorff condition. Let e_1 and e_2 be distinct members of E . Let $(f_k, g_k) \in e_k$, $k = 1, 2$. If

$$(C(e_1), V(e_1)) \neq (C(e_2), V(e_2)),$$

then on invoking the continuity at 0 of the component functions f_1, g_1, f_2, g_2 , we see that for small positive r the sets

$$\theta_{(f_k, g_k)}[\Delta(0; r)], \quad (2.6)$$

$k = 1, 2$, are disjoint. Suppose that

$$(C(e_1), V(e_1)) = (C(e_2), V(e_2)).$$

If f_1 is constant, then f_2 is not, and we see at once that the sets (2.6) are disjoint for small positive r . We proceed in the same way when any of the other three functions is constant. To continue, we suppose that f_1 and f_2 are not constant and that they have the canonical form of f_1 of Ex. 2.3. We then see that if the sets (2.6) are not disjoint for any small positive r , then by Lemma 2.1 we have $e_1 = e_2$. We conclude that the Hausdorff property holds.

We next examine the θ maps. Each is univalent because of the univalence

condition imposed on allowed (f, g) , and each is an open mapping by the very definition of T . The *continuity* of these maps may be established as follows. Suppose that (f_1, g_1) and (f_2, g_2) are allowed and that φ analytic at 0 satisfies $\varphi(0) = 0$ and

$$f_2(z) = f_1[\varphi(z)], \quad g_2(z) = g_1[\varphi(z)],$$

z small. On introducing τ_a as above relative to (f_1, g_1) and letting σ_b serve in the same capacity relative to (f_2, g_2) , we see that for each small b we have for z small the equality

$$f_2 \circ \sigma_b(z) = f_1 \circ \tau_{\varphi(b)}[\psi(z)]$$

where

$$\psi(z) = \tau_{\varphi(b)}^{-1}\{\varphi[\sigma_b(z)]\}.$$

A corresponding equality holds with g_k replacing f_k . Writing θ_k for $\theta_{(f_k, g_k)}$, θ for $\theta_{(f, g)}$, and θ^a for θ when (f, g) is replaced by $(f \circ \tau_a, g \circ \tau_a)$, we see that

$$\theta_2(z) = \theta_1[\varphi(z)],$$

z small, and

$$\theta^a = \theta \circ \tau_a.$$

These observations show that if $\theta_1(a) = \theta_2(b)$, there exists ψ analytic at a , $\psi(a) = b$, such that $\theta_1(z) = \theta_2[\psi(z)]$, z near a . Further ψ is locally determined near a by this condition. Suppose now that O is an open subset of E . We wish to show that $\theta^{-1}(O)$ is open in order to infer the continuity of θ . Since T is the family of unions of finite intersections of images with respect to the θ maps of open sets in their domains, when $\theta(a) \in O$, then

$$\theta(a) \in \theta_1(\Omega_1) \cap \cdots \cap \theta_n(\Omega_n) \subset O,$$

where each Ω_k is an open subset of the domain of θ_k . But from our observation concerning ψ above,

$$\theta(z) \in \theta_1(\Omega_1) \cap \cdots \cap \theta_n(\Omega_n),$$

z near a . Hence $\theta^{-1}(O)$ is open. The asserted continuity of θ follows.

Analytic entity. Let A be a component of E . Let $\Theta(A)$ be the set of θ maps with image in A . We show that $(A, \Theta(A))$ is a Riemann surface. Indeed, if $\theta_1, \theta_2 \in \Theta(A)$ and their images have a nonempty intersection O (necessarily open) and ψ is

$$\{(\theta_1^{-1}(p), \theta_2^{-1}(p)): p \in O\},$$

then we have

$$\theta_1(z) = \theta_2[\psi(z)],$$

$z \in \theta_1^{-1}(O)$, and from the developments of the preceding paragraph we see that ψ is analytic on its domain. Thus $\Theta(\mathbf{A})$ is a conformal structure on $\mathbf{A}$. By an *analytic entity* is meant a Riemann surface $(\mathbf{A}, \Theta(\mathbf{A}))$. It is immediate that center and value yield functions meromorphic on $\mathbf{A}$ (which we shall denote by C and V , respectively). For if $\theta = \theta_{(f,g)} \in \Theta(\mathbf{A})$, we have

$$C[\theta(z)] = f(z), \quad V[\theta(z)] = g(z),$$

z in the domain of θ .

EXERCISES

2.4. Suppose that for an allowed (f, g) the second component g is constant. Show that the value function whose domain is the analytic entity $\mathbf{A}$ containing $e(f, g)$ is constant and that there exists a univalent analytic map of $\hat{\mathbf{C}}$ onto $\mathbf{A}$. Hint: Let a denote the value taken by g and consider the map taking z into the element containing the ordered pair whose first component is the function $z + w$, $w \in \mathbf{C}$, and whose second component is the constant a , $z \in \mathbf{C}$, and taking ∞ into the element containing the ordered pair with first component $1/w$, domain $\mathbf{C}$, and second component the constant a .

2.5. Show that the map

$$\varphi = \{(e(f, g), e(g, f)): (f, g) \text{ allowed}\}$$

is a homeomorphism of $\mathbf{E}$ onto itself and that for each analytic entity $\mathbf{A}$ the map $\varphi \mid \mathbf{A}$ is a univalent analytic map of $\mathbf{A}$ onto an analytic entity. $\varphi(\mathbf{A})$ is termed the *inverse of A*. Justify the terminology.

Continuing the preparations for the proof of Theorem 2.1 we shall establish the following lemma relating ordered triples (S, f, g) where S is a Riemann surface and f and g are functions meromorphic on S , not both constant, with analytic entities.

Lemma 2.2: *Given (S, f, g) , there exists a unique $(\mathbf{A}, \varphi)$ where $\mathbf{A}$ is an analytic entity and φ is an analytic map of S into $\mathbf{A}$ such that $f = C \circ \varphi$, $g = V \circ \varphi$.*

PROOF: *Unicity.* Let $a \in S$ be such that the multiplicity of one of the functions f, g is 1 at a and let α be a uniformizer satisfying $\alpha(0) = a$ which is such that either $f \circ \alpha$ or $g \circ \alpha$ is univalent. Clearly $\varphi \circ \alpha$ is a uniformizer for $\mathbf{A}$. Further

$$e(f \circ \alpha, g \circ \alpha) = e(C \circ \varphi \circ \alpha, V \circ \varphi \circ \alpha) = \varphi(a), \quad (2.7)$$

as we see introducing a θ map, which we denote by θ , with image in $\mathbf{A}$ taking 0 into $\varphi(a)$, and noting that

$$C \circ \varphi \circ \alpha(z) = (C \circ \theta) \circ \theta^{-1}[\varphi \circ \alpha(z)], \quad (2.8)$$

and

$$V \circ \varphi \circ \alpha(z) = (V \circ \theta) \circ \theta^{-1}[\varphi \circ \alpha(z)], \quad (2.9)$$

z small. From (2.7) we infer the unicity of $\mathbf{A}$ which contains $e(f \circ \alpha, g \circ \alpha)$ and of φ given the arbitrary character of a . It is to be noted that the set where the multiplicity of a nonconstant meromorphic function on a Riemann surface is greater than one clusters at no point of the surface.

Existence. The case where f is constant is cared for by the analytic entity $\mathbf{A}$ for which the center function takes the constant value of f . There is exactly one $\mathbf{A}$ with this property. In this case the value function maps $\mathbf{A}$ univalently onto $\hat{\mathbb{C}}$ and $V^{-1} \circ g$ serves for φ .

Suppose then that f is not constant. Given $a \in S$, we assert that there exists a unique meromorphic element e such that, for every $(F, G) \in e$ and every uniformizer α for S , $\alpha(0) = a$, there exists h analytic at 0, $h(0) = 0$, such that

$$f \circ \alpha(z) = F[h(z)], \quad g \circ \alpha(z) = G[h(z)], \quad (2.10)$$

z small.

There is at most one such e . Let e_1 be a meromorphic element sharing the stated property with e , $(F_1, G_1) \in e_1$, and h_1 be analytic at 0, $h_1(0) = 0$, and such that the counterpart of (2.10) holds for z small when F, G, h are replaced by F_1, G_1, h_1 , respectively. By Ex. 2.3 and Lemma 2.1 we see that $e = e_1$.

There is such an e . We take a uniformizer α , $\alpha(0) = a$, with domain a disk $\Delta(0; r)$ so that $f \circ \alpha(z) = f(a) + z^{n(a; f)}$, when $f(a) \neq \infty$, and $f \circ \alpha(z) = 1/z^{n(a; f)}$, when $f(a) = \infty$, $|z| < r$, and $g \circ \alpha$ has a pole at most at 0. Let m denote the largest positive integer dividing $n(a; f)$ and each integer k for which $a_k \neq 0$ where (a_k) is the 0-Laurent sequence of $g \circ \alpha$. We take F and G meromorphic on $\Delta(0; \rho)$, where (i) $\rho \leq r^m$ when $r < +\infty$, (ii) $F(z) = f(a) + z^{n(a; f)/m}$ when $f(a) \neq \infty$, and $F(z) = 1/z^{n(a; f)/m}$ when $f(a) = \infty$, $|z| < \rho$, (iii) the k th 0-Laurent coefficient of G is a_{km} , $k \in \mathbb{Z}$, and (iv) (F, G) is allowed. We have

$$f \circ \alpha(z) = F(z^m), \quad g \circ \alpha(z) = G(z^m), \quad (2.11)$$

z small. It is now easy to verify that if $(F_1, G_1) \in e(F, G)$ and β is a uniformizer for S , $\beta(0) = a$, then there exists h analytic at 0, $h(0) = 0$, satisfying

$$f \circ \beta(z) = F_1[h(z)], \quad g \circ \beta(z) = G_1[h(z)],$$

z small. Hence $e(F, G)$ is such an e .

We shall produce $(\mathbf{A}, \varphi)$ satisfying the imposed conditions. Let $\varphi: S \rightarrow \mathbf{E}$ assign to $a \in S$ the unique meromorphic element satisfying the condition (2.10) relative to a . From (2.11) we see that

$$\varphi[\alpha(z)] = \theta_{(F, G)}(z^m), \quad (2.12)$$

z small. Hence not only is φ continuous on S —a fact which implies that $\varphi(S)$

is contained in an analytic entity $\mathbf{A}$ —but also φ is an analytic map of S into $\mathbf{A}$. Using (2.12) and thereupon (2.11) we conclude that $C \circ \varphi = f$, $V \circ \varphi = g$. The proof of Lemma 2.2 is complete.

EXERCISE

2.6. Let Ω denote the additive subgroup of $\mathbf{C}$ generated by $\{\omega_1, \omega_2\}$ where $\text{Im}(\overline{\omega_1}\omega_2) \neq 0$ and let p denote the associated Weierstrass p -function. Apply Lemma 2.2 to $(\mathbf{C}, p, p')$ and show that $\varphi(w) = \varphi(z)$ if and only if $w - z \in \Omega$. Show that $\mathbf{A}$ is compact but is not conformally equivalent to $\hat{\mathbf{C}}$. Obtain a necessary and sufficient condition for two such groups Ω to yield the same $\mathbf{A}$.

We are now in a position to treat Theorem 2.1. Given $\mathbf{W}$ and (S, f, g, φ) standing in the relation described by (2.1) to $\mathbf{W}$, we apply Lemma 2.2 to $(\mathbf{W}, c, v)$ obtaining $(\mathbf{A}_1, \psi_1)$ and to (S, f, g) obtaining $(\mathbf{A}_2, \psi_2)$. We conclude that $(\mathbf{A}_1, \psi_1) = (\mathbf{A}_2, \psi_2 \circ \varphi)$. Hence $\mathbf{A}_1 = \mathbf{A}_2$. Further ψ_1 and φ are univalent. Since $S - \varphi(\mathbf{W})$ clusters at no point of S , we conclude that ψ_2 is univalent. Thus $(\mathbf{A}_1, C, V, \psi_1)$ is an ordered quadruple fulfilling the conditions stated in the first sentence of Theorem 2.1. If (S, f, g, φ) fulfills the maximality condition, then Lemma 2.2 applied to (S, f, g) yields $(\mathbf{A}_1, \psi_2)$ and at the same time there exists an analytic map of $\mathbf{A}_1$ into S , say σ , such that $C = f \circ \sigma$, $V = g \circ \sigma$. It follows that $C = C \circ (\psi_2 \circ \sigma)$, $V = V \circ (\psi_2 \circ \sigma)$. Applying Lemma 2.2 we see that $\psi_2 \circ \sigma$ is the identity map of $\mathbf{A}_1$ onto itself. Hence ψ_2 has image $\mathbf{A}_1$. It is also easy to see that σ is a univalent map with image S on noting that $\sigma = \psi_2^{-1} \circ (\psi_2 \circ \sigma)$. The second assertion of Theorem 2.1 is now readily established.

The following remark is to be made. Suppose that Weierstrass classes $\mathbf{W}_1$ and $\mathbf{W}_2$ determine the same analytic entity. Then $\mathbf{W}_1 = \mathbf{W}_2$. Indeed their images in $\mathbf{A}$ have a point in common. The image of an element (a, P) , where P has domain $\Delta(a; r)$, is the meromorphic element containing (f, g) where f is $a + z$ with domain $\Delta(0; r)$ and g is $P(a + z)$ with domain $\Delta(0; r)$. Elements of $\mathbf{W}_1$ and $\mathbf{W}_2$ mapping into a given element of $\mathbf{A}$ are clearly equal. Hence $\mathbf{W}_1$ and $\mathbf{W}_2$ have a point in common. Our assertion follows.

EXERCISES

2.7. Let S be a Riemann surface and f and g functions meromorphic on S . We say that (S, f, g) is *extendible* provided that there exist a Riemann surface S_1 , functions f_1 and g_1 meromorphic on S_1 and an analytic map φ of S into S_1 which either is not univalent or else satisfies $\varphi(S) \neq S_1$ and at the same time satisfies $f = f_1 \circ \varphi$, $g = g_1 \circ \varphi$. Show that (S, f, g) is nonextendible if and only if φ of Lemma 2.2 maps S univalently onto $\mathbf{A}$.

2.8. Let W be a Weierstrass class. Let $a \in \hat{C}$, $r > 0$, $e_0 = (b, P) \in W$ where $b \in \Delta(a; r) - \{a\}$. It is supposed that (1) e_0 can be continued along every path in $\Delta(a; r) - \{a\}$ with initial point b ; (2) only a finite number of elements with center b are obtained in this manner, say m ; (3) there exists $\alpha \in \hat{C}$ such that for each positive ρ we have $v(e) \in \Delta(\alpha; \rho)$ when $c(e)$ is sufficiently near a , e being an element obtained by continuation from e_0 along a path in $\Delta(a; r) - \{a\}$; (4) $a = \infty$, $\alpha = \infty$, or $m > 1$. Show that there is a unique point $q \in A - \varphi(W)$ such that for ρ positive and small φ maps the set of e so obtained with centers in $\Delta(a; \rho) - \{a\}$ onto a "disk" punctured at q , that is, the image of a uniformizer θ with domain a disk centered at 0 less $\{\theta(0)\}$ and that $n(q; C) = m$. Show further that every point $q \in A - \varphi(W)$ is so associated with some (a, r, e_0) satisfying the above conditions. Of course, (A, φ) is understood in the sense of Lemma 2.2 relative to W . Hint: When $a \neq \infty$, let s satisfy $a + s^m = b$ and continue the element (s, P_1) , where $P_1(t) = P(a + t^m)$, t near s , along paths in $\Delta(0; r^{1/m}) - \{0\}$ issuing from s . One may proceed analogously when $a = \infty$, using t^{-m} in place of $a + t^m$.

2.9. Here $e = \exp(1)$. Let the element (e, P) be such that

$$P(e^{z^2}) = e^z,$$

z near 1, and let W be the Weierstrass class containing (e, P) . Study the set $A - \varphi(W)$ where (A, φ) is associated with (W, c, v) and show that there exists a univalent conformal map of W onto $C - \{0\}$.

3. COMPACT ANALYTIC ENTITIES AND RIEMANN SURFACES

On the basis of general mathematical experience we expect situations where compactness enters to be especially tractable. This is certainly true in the study of analytic entities as we shall now see.

Valence. The notion of *valence* for an arbitrary analytic map f of a Riemann surface S into a Riemann surface T is defined as earlier for meromorphic functions: the valence $v_f(q)$ at a point q of T is simply

$$\sum_{f(p)=q} n(p; f).$$

Suppose now that f is a nonconstant analytic map of a compact Riemann surface S into a Riemann surface T . We wish to study the valence of f . We first note that $v_f(q) < +\infty$, $q \in T$. Indeed, for each $p \in S$, there exists an open set $O(p)$ of S containing p such that f attains the point q in $O(p)$ at most at p . By the compactness of S we see that $S = \bigcup_{1 \leq k \leq n} O(p_k)$. Since

$$f^{-1}(\{q\}) \subset \{p_k: 1 \leq k \leq n\},$$

the finiteness of $v_f(q)$ follows. From the fact that $f(S)$ is open and compact, we conclude that $f(S) = T$. Given $q \in T$, let $p_1, \dots, p_m$ denote the points of $f^{-1}(\{q\})$, $p_j \neq p_k$ for $j \neq k$. Let O_k be a region containing p_k such that $v_{f|O_k}$

takes the value $n(p_k; f)$ at each point of $f(O_k)$, $O_j \cap O_k = \emptyset$ for $j \neq k$.[†] Since $S - \bigcup_{1 \leq k \leq n} O_k$ is compact and f omits q on it, its image with respect to f omits a neighborhood of q . Hence v_f takes the value $\sum_1^m n(p_k; f)$ at each point of some neighborhood of q . We conclude by the connectedness of T that v_f is a constant taking a positive integer as its value.

Suppose now that $\mathbf{A}$ is a compact analytic entity for which the center function is not constant. We let m denote the value taken by v_C . Since the multiplicity of C exceeds 1 only on a finite subset E of $\mathbf{A}$ (as we see by the argument used to prove that the valence of a nonconstant analytic map with domain a compact Riemann surface is finite-valued—here for each $e \in \mathbf{A}$ we introduce an open subset $O(e)$ of $\mathbf{A}$ containing e and such that the multiplicity of C is 1 at a point of $O(e) - \{e\}$), each point of $\hat{\mathbf{C}} - C(E)$ has exactly m preimages with respect to C .

We now show that F , the set of points $w \in \hat{\mathbf{C}}$ such that

$$\{V(e): C(e) = w\}$$

has fewer than m members, is finite. It suffices to show that F does not cluster at any point of $\hat{\mathbf{C}}$. Otherwise F would cluster at $w_0 \in \hat{\mathbf{C}}$ and there would exist $e_1, e_2 \in \mathbf{A}$ such that $C(e_1) = C(e_2) = w_0$, and sequences $(p_n), (q_n): \mathbf{N} \rightarrow \mathbf{A}$ such that $p_n \neq q_n$, $p_n \neq e_1$, $q_n \neq e_2$, $C(p_n) = C(q_n)$, $V(p_n) = V(q_n)$, $n \in \mathbf{N}$, and $\lim_{n \rightarrow \infty} p_n = e_1$, $\lim_{n \rightarrow \infty} q_n = e_2$. No assertion is made up to this point concerning the equality of e_1 and e_2 . By Lemma 2.1 of this chapter, we see that $e_1 = e_2$. However this is not possible since a map assigning $(C(e), V(e))$ to e whose domain is a suitably small neighborhood of e_1 is univalent. Our assertion follows.

The finiteness of F is conformally characteristic for compact analytic entities taken with their associated C and V , C not constant, in the class of (S, f, g) where S is a compact Riemann surface, f and g are meromorphic on S , and f is not constant. We shall prove the following theorem.

Theorem 3.1: *Given a compact Riemann surface S and functions f and g meromorphic on S , f not constant, then φ of the pair $(\mathbf{A}, \varphi)$ associated with (S, f, g) by Lemma 2.2 of this chapter is a univalent map of S onto $\mathbf{A}$ if and only if the set of $w \in \hat{\mathbf{C}}$ for which*

$$\{g(p): f(p) = w\}$$

has fewer members than the common value taken by v_f is finite. In fact, a

[†] It is to be observed that a region Ω of a Riemann surface is itself a Riemann surface when the conformal structure is just the set of inverses of the restrictions of the members φ of Φ (cf. §1) with domain ω_φ intersecting Ω to $\omega_\varphi \cap \Omega$. It is to be understood that regions of a Riemann surface are to be endowed in this manner with a conformal structure. $f|O_k$ is to be interpreted in these terms.

sufficient condition for φ to have the stated property is that the set of w in question be a proper subset of $\hat{\mathbb{C}}$.

PROOF: *Necessity* follows on noting that

$$\{g(p): f(p) = w\} = \{V[\varphi(p)]: C[\varphi(p)] = w\}.$$

Sufficiency. There exists $w \in \hat{\mathbb{C}}$ such that $f^{-1}(\{w\})$ has exactly m (the common value taken by v_f) members $p_1, \dots, p_m$ and $g(p_j) \neq g(p_k)$ when $j \neq k$. Clearly $n(p_k; f) = 1$, $k = 1, \dots, m$. From $f = C \circ \varphi$ we see that $n(p_k; \varphi) = 1$ $k = 1, \dots, m$. Further if $\varphi(p) = \varphi(p_1)$, then $f(p) = f(p_1)$; consequently, $p = p_k$ for some k . From $g(p_k) = g(p_1)$, we conclude $p = p_1$. Thus $v_\varphi(e) = 1$, $e \in \mathbf{A}$, and the proof of the sufficiency follows.

The problem of Riemann and Klein may be formulated as follows:

Given a compact Riemann surface S and f meromorphic on S and not constant, does there exist g meromorphic on S such that the component φ of $(\mathbf{A}, \varphi)$ associated with (S, f, g) by Lemma 2.2 of this chapter maps S univalently onto $\mathbf{A}$?

We have not established thus far that there exists a function meromorphic on S and not constant. We shall see that this is the case and that the problem of Riemann and Klein has a positive solution (cf. §§4–6). By virtue of Theorem 3.1 we may solve the problem of Riemann and Klein by taking $w \in \hat{\mathbb{C}}$ such that the number of members in $f^{-1}(\{w\})$ is precisely the common value taken by v_f and constructing g meromorphic on S taking distinct values at distinct points of $f^{-1}(\{w\})$ (cf. §4). An important consequence of the existence of a nonconstant function meromorphic on S and the solution in a positive sense of the problem of Riemann and Klein is the existence of an analytic entity conformally equivalent to S . Thus if we denote the class of Riemann surfaces conformally equivalent to S by $k(S)$, we see that the set of $k(\mathbf{A})$, $\mathbf{A}$ compact analytic entity, and the set of $k(S)$, S compact Riemann surface, are the same.

Formal polynomials. In order to continue our study of compact analytic entities we shall want some elementary facts concerning formal polynomials. It is to be expected that the reader is familiar with these matters (as is doubtless the case for most notions of an algebraic character employed in this book). However, for the sake of completeness we shall sketch the basic material we want. For this as well as other algebraic notions used by the worker in such areas of complex function theory we refer to [133].

In Ex. 1.1, Ch. XII, we introduced a ring A whose elements consisted of sequences $(a_n): \mathbb{N} \rightarrow F$ where F is a given field. The subset consisting of (a_n) satisfying $\sup\{n: a_n \neq 0\} < +\infty$ constitutes a subring $\mathfrak{P}$, termed the ring of *formal polynomials over F* . The fact that $\mathfrak{P}$ is a subring of A is readily checked.

Given $P = (a_n) \in \mathfrak{P}$, by the *degree of P* , denoted $d(P)$, is meant $\sup\{n: a_n \neq 0\}$. In particular, the degree of the zero element of $\mathfrak{P}$ is $-\infty$.

We have already encountered the notion of an ideal in Ch. VIII in studying the theorem of Bers; it was not important at that moment to insist on precise nomenclature. Here it will be appropriate to state some simple well-known facts in the standard language. Given a commutative ring $\mathfrak{R}$, $I \subset \mathfrak{R}$ is termed an *ideal of $\mathfrak{R}$* provided that I is a subgroup of $\mathfrak{R}$ considered as an additive group and $\cdot: \mathfrak{R} \times I \rightarrow I$. An ideal generated by a single element (that is, the intersection of the ideals containing the given element) is termed *principal*. For a ring $\mathfrak{R}$ with identity given $a \in \mathfrak{R}$ the ideal generated by $\{a\}$ is $\{ax: x \in \mathfrak{R}\}$. We shall denote it by $[a]$.

We shall want to have available the following two well-known results: (1) Given $P, Q \in \mathfrak{P}$, $d(P) \geq 0$, there exist unique $R, S \in \mathfrak{P}$ such that $Q = SP + R$, $d(R) < d(P)$. (Division algorithm.) (2) Every ideal in $\mathfrak{P}$ is principal.

(1) *Uniqueness* follows on noting that $d(P_1P_2) = d(P_1) + d(P_2)$. Since $SP + R = 0$ implies $S = 0$ and consequently $R = 0$, we readily conclude the asserted uniqueness.

Existence. If $0 \leq d(P) \leq d(Q)$, then $Q = CP + Q_1$ where $d(Q_1) < d(Q)$. In fact, with $P = (a_k)$, $Q = (b_k)$, and $l = d(Q) - d(P)$, C may be taken as (c_k) where $c_k = 0$ for $k \neq l$ and $c_l = b_{d(Q)}/a_{d(P)}$. We now proceed by induction. The case $d(Q) < d(P) + 0$ follows from the equality $Q = 0P + Q$. Suppose that the asserted representation holds when $d(Q) < d(P) + n$. When we consider P and Q such that $d(Q) = d(P) + n$, we have $Q = CP + Q_1$, $d(Q_1) < d(Q)$. Hence $d(Q_1) < d(P) + n$. Applying the induction hypothesis we obtain the desired representation.

(2) is now easy to conclude. Let I be an ideal in $\mathfrak{P}$. If $I = \{0\}$, the assertion is immediate. Otherwise let P_0 be a member of I having degree equal to the smallest of the degrees of the nonzero members of I . Given $P \in I$, we have $P = QP_0 + R$, $d(R) < d(P_0)$. Since $R \in I$, $R = 0$.

We say that $P \in \mathfrak{P}$ is *reducible* provided that there exist $P_1, P_2 \in \mathfrak{P}$, $d(P_k) < d(P)$, $k = 1, 2$, such that $P = P_1P_2$. Otherwise P is termed *irreducible*.

We term a polynomial $P = (a_k)$ of nonnegative degree d *normalized* provided that $a_d = 1$.

We are now ready to consider some algebraic aspects of analytic entities. Henceforth $\mathfrak{P}$ will denote the *ring of formal polynomials over the field $\mathbf{M}(\hat{\mathbf{C}})$* .† Given $P = (a_k) \in \mathfrak{P}$ and (S, f, g) , where S is a Riemann surface, f and g are meromorphic on S , and f is not constant, we say that (S, f, g) *annihilates P* provided that

$$\sum_{k=0}^{\infty} (a_k \circ f) g^k = 0.$$

† We recall that $\mathbf{M}(\hat{\mathbf{C}})$ is the set of functions meromorphic on $\hat{\mathbf{C}}$.

Our immediate objective is the following fundamental theorem concerning the relation between $\mathfrak{P}$ as now specialized and the set of compact analytic entities for which the center function is not constant.

Theorem 3.2: *There exists a unique map π from the set $\mathbf{K}$ of compact analytic entities for which the center function is not constant into the subset $\mathfrak{Q}$ of $\mathfrak{P}$ consisting of the normalized irreducible polynomials of positive degree such that $(\mathbf{A}, C, V)$ annihilates $\pi(\mathbf{A})$. The map π is univalent and has image $\mathfrak{Q}$.*

Further, if an analytic entity $\mathbf{A}$ (not assumed compact) with nonconstant center function is such that $(\mathbf{A}, C, V)$ annihilates a member of $\mathfrak{Q}$, then $\mathbf{A}$ is compact.

PROOF: *Uniqueness.* Let $\mathbf{A}$ be a given analytic entity such that $(\mathbf{A}, C, V)$ annihilates some member of $\mathfrak{Q}$. Let P be a member of $\mathfrak{P}$ annihilated by $(\mathbf{A}, C, V)$ of least positive degree and normalized. Let Q be a member of $\mathfrak{Q}$ annihilated by $(\mathbf{A}, C, V)$. By the division algorithm $Q = SP + R$, where $S, R \in \mathfrak{P}$, and $d(R) < d(P)$. Since $(\mathbf{A}, C, V)$ annihilates R , we see that $R = 0$. Hence there is just one member of $\mathfrak{Q}$ annihilated by $(\mathbf{A}, C, V)$, namely P . The uniqueness of π follows. It is to be noted that thus far the compactness of $\mathbf{A}$ has not been invoked.

Existence. Given $\mathbf{A} \in \mathbf{K}$, we introduce $n + 1$, the common value taken by v_C , and for each z of $\hat{\mathbf{C}}$ in the complement of the image with respect to C of

$$\{e: n(e; C) > 1\} \cup \{e: V(e) = \infty\}$$

we define $a_0(z), \dots, a_n(z)$ by the stipulation

$$w^{n+1} + \sum_0^n a_k(z) w^k = \prod_{C(e)=z} [w - V(e)], \quad w \in \mathbf{C}. \quad (3.1)$$

It is to be observed that the a_k define functions analytic on the set of admitted z . To see this, it suffices to consider an admitted z , say b , to introduce for each $e \in C^{-1}(\{b\})$ a neighborhood $U(e)$ of e on which C is univalent, thereupon to introduce for each e the restriction φ_e of the inverse of the $C|U(e)$ to $\bigcap_{C(e)=b} C[U(e)]$, and to observe that for z in this last set sufficiently near b the right-hand side of (3.1) is equal to

$$\prod_{C(e)=b} \{w - V[\varphi_e(z)]\}. \quad (3.2)$$

The asserted analyticity follows.

We now show that each function a_k has at worst poles at the points z that were not admitted and thus is the restriction of a member of $\mathbf{M}(\hat{\mathbf{C}})$, which we shall denote also by a_k . Let b be a nonadmitted point. Let m denote a Möbius transformation satisfying $m(b) = 0$ and take μ a positive integer so

that for each $e \in C^{-1}(\{b\})$ the multiplicity of $m^\mu \circ C$ at e is at least as large as $-\partial_V(e)$. For z near b but different from it, we obtain from the equality

$$w^{n+1} + \sum_0^n [m(z)]^{\mu(n+1-k)} a_k(z) w^k = \prod_{C(e)=z} [w - m^\mu \circ C(e) V(e)]$$

and the fact that $(m^\mu \circ C)V$ is analytic at each point of $C^{-1}(\{b\})$ the conclusion that $m(z)^{\mu(n+1-k)} a_k(z)$ has a removable singularity at b . Hence a_k has at worst a pole at b .

We propose P with $a_0, \dots, a_n$ as given, $a_{n+1} = 1$, $a_{n+2+k} = 0$, $k \in \mathbb{N}$. We assert that $(\mathbf{A}, C, V)$ annihilates P and that P is irreducible. Indeed, taking e_0 such that $C(e_0)$ is an allowed z we see from (3.1) that

$$[V(e)]^{n+1} + \sum_0^n a_k[C(e)][V(e)]^k = 0,$$

e near e_0 , and we conclude that $(\mathbf{A}, C, V)$ annihilates P . Suppose now that $Q = (b_k)$ is a normalized member of $\mathfrak{P}$ of least positive degree which is annihilated by $(\mathbf{A}, C, V)$. We see that

$$V^{m+1} + \sum_0^m (b_k \circ C) V^k = 0,$$

where $m+1 = d(Q)$. Applying the fact established earlier that $\{V(e): C(e) = z\}$ has $n+1$ members save for a finite set of z , we see that we may take z such that $\{V(e): C(e) = z\}$ has $n+1$ members, all different from ∞ . Since

$$[V(e)]^{m+1} + \sum_0^m b_k(z) [V(e)]^k = 0$$

for $e \in C^{-1}(\{z\})$, we conclude that $m \geq n$ and hence $m = n$. Using the minimal property of $d(Q)$ and the division algorithm we conclude that $P = Q$. If P were reducible, then P would be representable as the product of two members of $\mathfrak{P}$ each having a smaller degree than P ; but $(\mathbf{A}, C, V)$ would annihilate one of these factors and the minimal property of $d(P)$ would be violated. Hence P is irreducible.

The existence of a map π is thereby established.

$\pi(\mathbf{K}) = \mathfrak{Q}$. Given $P = (a_k) \in \mathfrak{Q}$, $d(P) = n+1$, we introduce

$$P' = ((k+1)a_{k+1}),$$

the formal derivative of P . The ideal in $\mathfrak{P}$ generated by $\{P, P'\}$ is principal, say $[Q]$, and clearly $d(Q) \geq 0$, since there are nonzero members. Since P is irreducible, $d(Q) = 0$, and hence $[Q] = \mathfrak{P}$. Since the ideal generated by $\{P, P'\}$ consists of the polynomials of the form $AP + BP'$, A and B in $\mathfrak{P}$, we

conclude that, for some A and B ,

$$AP + BP' = 1. \dagger \quad (3.3)$$

Let Ω denote the set of $z \in \mathbb{C}$ such that the components of A , B , and P are all analytic at z . The complement $\hat{\mathbb{C}} - \Omega$ is finite. The set of roots of

$$w^{n+1} + \sum_0^n a_k(z) w^k = 0 \quad (3.4)$$

has exactly $n + 1$ members for each $z \in \Omega$ as a consequence of (3.3).

We may control the size of the roots of (3.4) for z near a point $b \in \hat{\mathbb{C}} - \Omega$. Indeed, introducing as above a Möbius transformation m , $m(b) = 0$, we see that $m^{\mu(n+1-k)} a_k$ takes the value 0 at b , $k = 0, \dots, n$, when μ is a suitably chosen positive integer. By the theorem of Rouché the roots of

$$w^{n+1} + \sum_0^n [m(z)]^{\mu(n+1-k)} a_k(z) w^k = 0 \quad (3.5)$$

are all arbitrarily small when $z \in \Omega$ is sufficiently near b . But for $z \in \Omega$ near b the roots of this equation are simply the roots of (3.4) multiplied by $[m(z)]^\mu$. We shall use this observation in the next paragraph.

We fix $z_0 \in \Omega$ and conclude, using the local existence theorem for algebroid equations, Ex. 9.9, Ch. V, the existence of an element $(z_0, g) \in \mathfrak{E}$ satisfying

$$[g(z)]^{n+1} + \sum_0^n a_k(z) [g(z)]^k = 0, \quad (3.6)$$

z near z_0 . The first part of the following argument will proceed in the milieu of a Weierstrass class and its conclusion will be translated into the terms of the associated analytic entity. We consider an arbitrary path γ in Ω with initial point z_0 and show that (z_0, g) admits continuation along γ . The argument is a standard one and proceeds by considering the supremum of an appropriate set of real numbers. We take $[0, 1]$ as the domain of γ and let τ be the supremum of the numbers t in $]0, 1]$ such that (z_0, g) admits continuation along $\gamma| [0, t]$. Clearly, $0 < \tau \leq 1$. Let κ denote the union of such continuations. The domain of κ is $[0, \tau]$ or $[0, \tau[$. Now $\lim_t v \circ \kappa$ exists and is one of the roots of (3.4) with $z = \gamma(\tau)$ as we see by appeal to the connectedness of a semi-interval and the continuity of v . The element $(\gamma(\tau), h)$, where $h[\gamma(\tau)] = \lim_t v \circ \kappa$ and h satisfies (3.6) replacing g for z near $\gamma(\tau)$, has the property that $\kappa(t)$ is a direct continuation of it for t near τ . Hence $\tau = 1$. For otherwise, by defining $\kappa_1(s)$ as $\kappa(s)$, $0 \leq s < \tau$, and as the direct continuation of $(\gamma(\tau), h)$ for s near τ , we would conclude the existence of an allowed t greater than τ . Further the

† Here $1 =$ the identity of the ring $\mathfrak{F}$.

domain of κ is $[0, 1]$ as we see on reference to κ_1 . Thus (z_0, g) admits continuation along γ .

We are led to the following conclusion: m , the number of elements with center $z \in \Omega$ which are continuations of (z_0, g) along paths in Ω , is independent of z and does not exceed $n + 1$.

By a paraphrase of our earlier arguments concerning the existence of a polynomial annihilated by a compact analytic entity with nonconstant center function we see that the coefficients b_k specified by

$$w^m + \sum_0^{m-1} b_k(z) w^k = \prod [w - v(e)],$$

$w \in \mathbb{C}$, where $z \in \Omega$ and the product is taken over the elements e with center $z \in \Omega$ which are continuations of (z_0, g) along paths in Ω , are analytic on Ω . By the observation concerning the roots of (3.5) we see that the coefficients b_k have removable singularities or poles at the points of $\hat{C} - \Omega$. We denote the member of $\mathbf{M}(\hat{C})$ whose restriction to Ω is b_k also by b_k and introduce the polynomial Q whose k th component is b_k , $k = 0, \dots, m-1$, whose m th component is 1, and whose others are 0. The ideal generated by $\{P, Q\}$ is principal. Let it be generated by R . Now $d(R) = m$ since, if $R = (c_k)$, we have $\sum_0^\infty c_k(z) w^k = 0$ when

$$w^m + \sum_0^{m-1} b_k(z) w^k = 0$$

save possibly for a finite set of z in Ω , and also $0 \leq d(R) \leq m$. Hence the ideal generated by $\{P, Q\}$ is generated by Q and we conclude that $P = Q$.

Now let $\mathbf{W}$ denote the Weierstrass class containing (z_0, g) . We note that, for each $z \in \Omega$, $c^{-1}(\{z\})$ contains exactly $n + 1$ elements (c having domain $\mathbf{W}$). Indeed, this follows since $c^{-1}(C - \Omega)$ clusters at no point of $\mathbf{W}$ and hence $c^{-1}(\Omega)$ is arcwise connected.

Let $\mathbf{A}$ be the analytic entity and φ the univalent analytic map of $\mathbf{W}$ into $\mathbf{A}$ such that $c = C \circ \varphi$, $v = V \circ \varphi$. Since $\mathbf{A} - \varphi(\mathbf{W})$ clusters at no point of $\mathbf{A}$, the valence of the center function C for $\mathbf{A}$ does not exceed $n + 1$ anywhere. It takes the value $n + 1$ at each point of Ω . We wish to show that v_c takes the value $n + 1$ at each point $\zeta \in \hat{C} - \Omega$. We introduce a disk $\Delta(\zeta; r)$, $r < +\infty$, such that $\Delta(\zeta; r) - \{\zeta\} \subset \Omega$ and note that if e is a given element of $\mathbf{W}$ with $c(e) \in \Delta(\zeta; r) - \{\zeta\}$, then e may be continued along all paths in $\Delta(\zeta; r) - \{\zeta\}$ issuing from $c(e)$. Let $W(e)$ denote the set of elements which are obtained from e by continuation along such paths, that is, the union of the images of such continuations. If $e_1, e_2 \in \mathbf{W}$, and $c(e_1), c(e_2) \in \Delta(\zeta; r) - \{\zeta\}$, then either $W(e_1) = W(e_2)$ or else $W(e_1) \cap W(e_2) = \emptyset$. Let $W_1, \dots, W_m$ denote the $W(e)$, $W_j \neq W_k$ when $j \neq k$. The number, μ_k , of elements contained in W_k and

having a given center in $\Delta(\zeta; r) - \{\zeta\}$ is independent of the center. Further $\sum_1^m \mu_k = n + 1$. By the argument of Ex. 2.8, this chapter, there exists a point $q_k \in \mathbf{A}$ such that φ maps W_k onto $\theta[\Delta(0; 1)] - \{q_k\}$ where θ is some uniformizer for $\mathbf{A}$ with domain $\Delta(0; 1)$ satisfying $\theta(0) = q_k$. It is to be noted that the condition (3) of the cited exercise is indeed fulfilled. This may be verified with the aid of the observation that if $\lim_{n \rightarrow \infty} v(e_n) = \alpha \in \mathbf{C}$ where $e_n \in W_j$, $\lim_{n \rightarrow \infty} c(e_n) = \zeta$, then α is a root of $\sum_0^\infty b_k w^k = 0$, b_k being the value at ζ of the k th component of λP where $d(\lambda) = 0$ and the divisor at ζ of the zeroth component of λ is $\max\{-\partial_{a_k}(\zeta): k \in \mathbf{N}\}$. Here $P = (a_k)$. If (3) is not fulfilled, the set of such limit values α is infinite.[†] We have $n(q_k; C) = \mu_k$ but in contrast to the situation of that exercise we must reckon with the possibility: $\mu_k = 1$, $C(q_k) \neq \infty$, $V(q_k) \neq \infty$. Further $q_j \neq q_k$ when $j \neq k$, otherwise $W_j \cap W_k \neq \emptyset$. We conclude that $v_C(\zeta) = n + 1$. Thus v_C is constant.

We observe that the following general result holds: *If there exists an analytic map f of a Riemann surface S into a compact Riemann surface T such that v_f is a finite-valued constant, then S is compact.* We may see this as follows. Given $q \in T$, there exists an open neighborhood $O(q)$ of q such that $f^{-1}[O(q)]$ is relatively compact, for a relatively compact open subset of S containing $f^{-1}(\{q\})$ contains all the preimages with respect to f of points sufficiently near q . To see this it suffices to consider the behavior of f on small disjoint neighborhoods of the points of $f^{-1}(\{q\})$. By the compactness of T there exists a finite set of such $O(q)$ with union T . The union of their preimages is S and the compactness of S follows.

Applying this result to $\mathbf{A}$, we see that $\mathbf{A}$ is compact.

Using the fact that (3.6) holds we see that $(\mathbf{A}, C, V)$ annihilates P . In other words $\pi(\mathbf{K}) = \Omega$.

Suppose now that $\mathbf{B}$ is an analytic entity with nonconstant C such that $(\mathbf{B}, C, V)$ annihilates P . We let $q_0 \in \mathbf{B}$ be such that $C(q_0) \in \Omega$ and $V(q_0) \neq \infty$. If $(C(q_0), g) \in \mathfrak{E}$ is such that $g[C(q_0)] = V(q_0)$ and g satisfies (3.6) for z near $C(q_0)$, we have $g[C(q)] = V(q)$ for q near q_0 . Now $(C(q_0), g) \in \mathbf{W}$. Hence $q_0 \in \mathbf{A}$ and consequently $\mathbf{A} = \mathbf{B}$. We conclude that π is univalent. It is to be noted that $\mathbf{B}$ is not assumed to be compact.

Theorem 3.2 follows.

We now turn to an examination of the algebraic structure of $\mathbf{M}(\mathbf{A})$ where $\mathbf{A}$ is a compact analytic entity with nonconstant center function. Our object is to show that $\mathbf{M}(\mathbf{A})$ is isomorphic to a field constructed in a purely algebraic way from $\mathfrak{P}$ and $Q = \pi(\mathbf{A})$. We shall want to review some simple algebraic facts before proceeding. We introduce an equivalence relation in $\mathfrak{P}$ by the stipulation that P_2 is to be equivalent to P_1 provided that $P_2 - P_1$ is in

[†] This part of Theorem 3.2 can be treated elegantly with the aid of information concerning the covering maps associated with a punctured disk, but the exposition would take us outside of the program of this book.

$[Q]$. We have a bona fide equivalence relation between members of $\mathfrak{P}$. Let $k(P)$ denote the equivalence class containing P . We introduce $\mathfrak{P}/[Q]$, the set of $k(P)$. We define an *addition* on $\mathfrak{P}/[Q]$ by

$$k(P_1) + k(P_2) = k(P_1 + P_2),$$

and a *multiplication* by

$$k(P_1) k(P_2) = k(P_1 P_2).$$

The operations so defined are functions on the cartesian product of $\mathfrak{P}/[Q]$ with itself and $\mathfrak{P}/[Q]$ taken with these operations constitutes a field. These assertions are easily verified, and, of course, are well known. The main point to be noted is that if P does not belong to $[Q]$ there exist $A, B \in \mathfrak{P}$ such that $AP + BQ = 1$. We have $k(A) k(P) = k(1)$, which is the identity element of $\mathfrak{P}/[Q]$.

We now define a map $\varphi: \mathfrak{P}/[Q] \rightarrow \mathbf{M}(\mathbf{A})$ by the stipulation

$$\varphi = \{(k(P), h_P): P \in \mathfrak{P}\}$$

where

$$h_P = \sum_0^{\infty} (b_k \circ C) V^k,$$

$P = (b_k) \in \mathfrak{P}$, and assert that φ is an isomorphism of $\mathfrak{P}/[Q]$ onto $\mathbf{M}(\mathbf{A})$. Here C and V are the center and value functions on $\mathbf{A}$.

We note that φ is a *function*, for if P_1 and P_2 are equivalent members of $\mathfrak{P}$, $h_{P_2} = h_{P_1}$. If $h_{P_1} = h_{P_2}$, then $h_{P_2 - P_1} = 0$. Now if $h_P = 0$, we see on applying the division algorithm that P is in $[Q]$, for otherwise $(\mathbf{A}, C, V)$ would annihilate an irreducible polynomial of positive degree less than $d(Q)$. We conclude that φ is *univalent*. Since $h_{P_1} + h_{P_2} = h_{P_1 + P_2}$ and $h_{P_1} h_{P_2} = h_{P_1 P_2}$, we see that φ preserves addition and multiplication.

There remains to be verified that $\mathbf{M}(\mathbf{A}) = \varphi(\mathfrak{P}/[Q])$. This assertion will follow on showing that if $f \in \mathbf{M}(\mathbf{A})$, there exist $b_0, \dots, b_n \in \mathbf{M}(\hat{\mathbf{C}})$ such that

$$f = \sum_0^n (b_k \circ C) V^k.$$

Here $n + 1 = d(Q)$. We take Ω as the set of z in $\mathbf{C}$ such that $\{p: C(p) = z\}$ has $n + 1$ members at each of which f and V take values in $\mathbf{C}$ and such that $\{V(p): C(p) = z\}$ has $n + 1$ members. We have

$$\sum_{C(q)=z} \frac{f(q)}{w - V(q)} = \left[\sum_0^n d_k(z) w^k \right] / \left[w^{n+1} + \sum_0^n a_k(z) w^k \right]. \quad (3.7)$$

Here $(a_k) = Q$ and $d_0, \dots, d_n$ are analytic on Ω . Indeed, given q such that

$C(q) = z$ we see that $f(q)$ is the residue at $V(q)$ of the rational function defined by each side of (3.7), and hence

$$h_{Q'}(q) f(q) = \sum_0^n d_k[C(q)][V(q)]^k. \quad (3.8)$$

Here Q' is the formal derivative of Q . Now let $q_1, \dots, q_{n+1}$ denote the members of $C^{-1}(\{z_0\})$ where $z_0 \in \Omega$ and let φ_k denote the inverse of C restricted to a neighborhood of q_k on which C is univalent. For z near z_0 using the ∞ -Laurent coefficients of

$$\sum_{C(q)=z} \frac{f(q)}{w - V(q)} = \sum_1^{n+1} \frac{f[\varphi_k(z)]}{w - V[\varphi_k(z)]},$$

we represent the $d_k(z)$ in terms of these coefficients and the $a_k(z)$, and conclude that $d_0, \dots, d_n$ are analytic at z_0 and consequently on Ω . The proof will be complete once we show that the d_k have removable singularities or poles at the points of $\hat{\mathbb{C}} - \Omega$, for $k(Q')$ has an inverse so that there exists $S \in \mathfrak{P}$ such that $h_S h_{Q'} = 1$ and by (3.8) we have

$$f = h_S \sum_0^n (d_k \circ C) V^k = h_T$$

for some $T \in \mathfrak{P}$.†

To treat the behavior of the d_k at a point $\zeta \in \hat{\mathbb{C}} - \Omega$, we use again the now familiar device of introducing an auxiliary factor with the aid of a Möbius transformation m satisfying $m(\zeta) = 0$. We take μ a positive integer such that $(m^\mu \circ C)f$ and $(m^\mu \circ C)V$ are analytic at each point of $C^{-1}(\{\zeta\})$. For z in $\Delta(\zeta; r) - \{\zeta\}$, r small, we have

$$\sum_{C(q)=z} \frac{[m(z)]^\mu f(q)}{w - [m(z)]^\mu V(q)} = \left[\sum_0^n \tilde{d}_k(z) w^k \right] / \left[w^{n+1} + \sum_0^n \tilde{a}_k(z) w^k \right]$$

where

$$\tilde{a}_k(z) = [m(z)]^{\mu(n+1-k)} a_k(z)$$

and

$$\tilde{d}_k(z) = [m(z)]^{\mu(n+1-k)} d_k(z).$$

Now the $\tilde{d}_k$ are analytic in $\Delta(\zeta; r) - \{\zeta\}$. They have removable singularities at ζ , as we see using ∞ -Laurent coefficients once more. The asserted behavior of the d_k follows.

† Here the d_k are construed in the obvious way as members of $\mathbf{M}(\hat{\mathbb{C}})$.

EXERCISES

3.1. Suppose that $a_0, \dots, a_n, n \in \mathbb{N}$, are analytic on an annulus $\{r < |z| < s\}$, $0 \leq r < s \leq +\infty$, and that for each z in the annulus $w^{n+1} + \sum_0^n a_k(z) w^k = 0$ has $n+1$ distinct roots. Show that there exists a positive integer μ such that for $w \in \mathbb{C}$, $r^{1/\mu} < |z| < s^{1/\mu}$, we have

$$w^{n+1} + \sum_0^n a_k(z^n) w^k = \prod_1^{n+1} [w - b_k(z)],$$

where $b_1, \dots, b_{n+1}$ are analytic on $\{r^{1/\mu} < |z| < s^{1/\mu}\}$. Show that $(n+1)!$ is an allowable choice of μ .

3.2. Develop the theory of formal polynomials over $M(A)$, the field of functions meromorphic on a region $A \subset \hat{\mathbb{C}}$. Show for such a polynomial of positive degree which is irreducible the existence of an annihilating triple (S, f, g) where $f(S) = A$ and $\nu_f(z) = n+1$, $z \in A$, $n+1$ being the degree of the polynomial.

4. THE PROBLEM OF RIEMANN AND KLEIN

In §§5 and 6 we shall establish the existence of a function meromorphic on a Riemann surface S and having the property that it takes distinct values at points $a, b \in S$ where $a \neq b$. Such a function is said to *distinguish* a and b . In the present section we shall explore the consequences of the existence for given distinct $a, b \in S$ of functions meromorphic on S which distinguish a and b . Trivially, we are assured of the existence of functions meromorphic on S which are not constant. But we have more: given a compact Riemann surface S and f meromorphic on S and not constant, we may construct g meromorphic on S (and nonconstant) such that (S, f, g) satisfies the criterion of Theorem 3.1 of this chapter. The problem of Riemann and Klein is thereby solved.

Construction of g . Let μ denote the common value of the valence of f . If $\mu = 1$, we may take $g = f$. Suppose that $\mu > 1$ and let $w \in \hat{\mathbb{C}}$ be such that $f^{-1}(\{w\})$ contains exactly μ members. Given a, b distinct and satisfying $f(a) = f(b) = w$, by our assumption concerning meromorphic distinguishing functions there exists h meromorphic on S satisfying $h(a) \neq h(b)$. Let m denote a Möbius transformation satisfying $m[h(a)] = 1$, $m[h(b)] = 0$, and $m(\alpha) = \infty$ where α is some point of $\hat{\mathbb{C}} - \{h(p) : f(p) = w\}$. The function $h_{a,b} = m \circ h$ takes finite values at the points of $f^{-1}(\{w\})$, the value 1 at a and the value 0 at b . On taking the product of functions $h_{a,b}$, a fixed and b ranging over the remaining points of $f^{-1}(\{w\})$, we obtain a function h_a meromorphic on S taking the value 1 at a and the value 0 at the remaining points of $f^{-1}(\{w\})$. If

λ is a univalent map of $f^{-1}(\{w\})$ into $\mathbb{C}$, then

$$\sum_{a \in f^{-1}(\{w\})} \lambda(a) h_a$$

is an admitted g .

5. DISTINGUISHING MEROMORPHIC FUNCTIONS

We propose to reduce the problem of the existence of distinguishing meromorphic functions to a problem concerning the existence of a harmonic function with a singularity (of a nature to be specified) at a given point of a Riemann surface. The latter problem will be solved with the aid of Perron methods in §6. Our first step will be the introduction of the notion of an *abelian* (or *meromorphic*) *differential on a Riemann surface*. The concept is of basic importance throughout the theory of Riemann surfaces.

Let Θ be the conformal structure on a Riemann surface S . By an *abelian differential on S* is meant a map ω with domain Θ such that (1) $\omega(\theta)$, the image with respect to ω of $\theta \in \Theta$, is a function meromorphic on the domain of θ , (2) if the images of members θ_1 and θ_2 of Θ have a nonempty intersection O and if

$$\alpha = \{(\theta_1^{-1}(p), \theta_2^{-1}(p)) : p \in O\},$$

and $A = \theta_1^{-1}(O)$, then

$$\omega(\theta_1)|_A = [\omega(\theta_2) \circ \alpha]\alpha'. \quad (5.1)$$

An example of special interest is the *differential, df , of a function f meromorphic on S* given by the stipulation $df(\theta) = (f \circ \theta)'$, $\theta \in \Theta$. The relation (5.1) is seen to hold for df by virtue of the equality

$$(f \circ \theta_1)|_A = (f \circ \theta_2) \circ \alpha$$

and the chain rule.

It is simple to establish the following properties of abelian differentials.

(1) *Given an abelian differential ω on S , either every $\omega(\theta)$ is the constant zero or none is.*

To see this we note that if $\omega(\theta_1)$ is the constant zero and $\omega(\theta_2)$ is not, then the images of θ_1 and θ_2 are disjoint. From the connectedness of S only one of the asserted possibilities can occur, for otherwise S would admit a splitting. When every $\omega(\theta)$ is the constant zero, we term ω the *zero differential*.

(2) *Given an abelian differential ω on S and $\theta_1, \theta_2 \in \Theta$ whose images have a point q in common, then the divisor of $\omega(\theta_1)$ at $\theta_1^{-1}(q)$ is equal to the divisor of $\omega(\theta_2)$ at $\theta_2^{-1}(q)$.*

We define $\partial(q; \omega)$, the *divisor of ω at $q \in S$* , as the common value of the divisor of $\omega(\theta)$ at $\theta^{-1}(q)$ where $\theta \in \Theta$ has an image containing q .

(3) Given ω_1 and ω_2 , abelian differentials on S , ω_2 being not the zero differential. Then

$$f = \bigcup_{\theta \in \Theta} \left[\frac{\omega_1(\theta)}{\omega_2(\theta)} \right] \circ \theta^{-1} \quad (5.2)$$

is a function meromorphic on S satisfying

$$\partial_f(q) = \partial(q; \omega_1) - \partial(q; \omega_2), \quad q \in S.$$

The verification is immediate on referring to (5.1).† We denote f by ω_1/ω_2 and term it the *quotient*, ω_1 divided by ω_2 .

(4) Given a, b distinct points of S and ω_1 and ω_2 , abelian differentials on S satisfying $\partial(a; \omega_1) < 0 \leq \partial(b; \omega_1)$ and $\partial(b; \omega_2) < 0 \leq \partial(a; \omega_2)$, then ω_1/ω_2 distinguishes a and b .

(5) Suppose that $a \in S$, that u is harmonic on $S - \{a\}$, and that in terms of a uniformizer σ , $\sigma(0) = a$, $u[\sigma(z)]$ admits for z small and not 0 a representation of the form

$$c \log |z| + \operatorname{Re} f(z)$$

where $c \in \mathbf{R}$ and f has a pole of order n at 0. Then ω defined by

$$\omega(\theta) = (u \circ \theta)_z,$$

the value being by convention ∞ at $\theta^{-1}(a)$ for θ with image containing a , is an abelian differential on S satisfying $\partial(a; \omega) = -(n+1)$, $\partial(p; \omega) \geq 0$, $p \neq a$.

A more detailed examination of abelian differentials is, of course, essential for a full-scale study of Riemann surface theory. We have just isolated some properties that we shall want. It is now clear that if for arbitrary $a \in S$ there exists a harmonic function u with the stated behavior, then for given distinct $a, b \in S$ we are led to differentials ω_1, ω_2 such that ω_1/ω_2 distinguishes a and b . Here (3), (4), and (5) are relevant.

The following exercises, which are not needed in the remaining sections of the book, pursue the subject of abelian differentials further and are offered for the interested reader.

EXERCISES

5.1. Given ω , an abelian differential on S , $a \in S$, lying in the images of $\theta_1, \theta_2 \in \Theta$. Show that the residue of $\omega(\theta_1)$ at $\theta_1^{-1}(a)$ is equal to the residue of $\omega(\theta_2)$ at $\theta_2^{-1}(a)$.

The common value of the residue of $\omega(\theta)$ at $\theta^{-1}(a)$ for $\theta \in \Theta$ whose images contain

† Here by convention $(+\infty) - a = +\infty$ when $a \in \mathbf{R}$.

a is termed the *residue of ω at a* and is denoted $\text{res}(a; \omega)$. Relate the notion of the residue of a meromorphic function on a region $A \subset \hat{\mathbb{C}}$ at a point of A with the notion of the residue of an abelian differential at the point by introducing (1) Θ as the set consisting of θ_1 , the identity map on $A \cap \mathbb{C}$, and θ_2, z^{-1} with domain the image of $A - \{0\}$ with respect to z^{-1} , (2) $\omega(\theta_1) = f|_{(A \cap \mathbb{C})}$ and $\omega(\theta_2) = (f \circ \theta_2)\theta_2'$ (cf. §3, Ch. VI).

5.2. Given ω as in Ex. 5.1 and f meromorphic on S , the map assigning to $\theta \in \Theta$ the function $(f \circ \theta)\omega(\theta)$ is an abelian differential on S . It is denoted $f\omega$. If ω is not the zero differential and ω_1 is an abelian differential on S , then $\omega_1 = (\omega_1/\omega)\omega$.

5.3. Given a compact Riemann surface S , the sum

$$\sum_{p \in S} \partial(p; \omega)$$

is the same for all abelian differentials ω on S which are not the zero differential. The value in question will be denoted by $\chi(S)$. [$\chi(S)$ has topological significance. It is equal to the Euler–Poincaré characteristic of S , if we follow one of the two definitions given for the Euler–Poincaré characteristic, and otherwise it is (-1) times the Euler–Poincaré characteristic of S .] Show that $\chi(\hat{\mathbb{C}}) = -2$ by calculating the sum in question for the differential of the identity map. Determine $\chi(A)$ for the analytic entity A associated with (C, p, p') in the sense of Lemma 2.2 of this chapter, p being a p -function.

5.4. The *sum of abelian differentials on S* , say ω_1 and ω_2 , is defined as the map assigning $\omega_1(\theta) + \omega_2(\theta)$ to $\theta \in \Theta$. It is evidently an abelian differential on S . We denote it by $\omega_1 + \omega_2$. Show that with addition so defined and $c\omega, c \in \mathbb{C}$, defined as the map assigning $c\omega(\theta)$ to θ , the set of abelian differentials on S is a vector space over $\mathbb{C}$.

***5.5.** Let S be a compact Riemann surface and let $\alpha: S \rightarrow \mathbb{Z}$ be such that $\{p: \alpha(p) \neq 0\}$ is finite. Show that $\mathfrak{B}_1(\alpha)$, the set of functions f meromorphic on S which satisfy

$$\partial f(p) + \alpha(p) \geq 0, \quad p \in S,$$

together with addition and multiplication by scalars as defined for meromorphic functions, is a finite-dimensional vector space over $\mathbb{C}$. Show that a corresponding statement holds for $\mathfrak{B}_2(\alpha)$, the set of abelian differentials on S satisfying

$$\partial(p; \omega) + \alpha(p) \geq 0, \quad p \in S,$$

the definitions of Ex. 5.4 prevailing.

Hint: For meromorphic functions consider $\mathfrak{B}_1(\alpha^+)$ and examine the Laurent expansions of $f \in \mathfrak{B}_1(\alpha^+)$ composed with a uniformizer $\sigma, \sigma(0) = a$, for a satisfying $\alpha(a) > 0$. With $(c_k(f))$ the 0-Laurent sequence of $f \circ \sigma$ consider the null space and rank of the linear transformation mapping f into $(c_{-k}(f))_{k \in \mathbb{N}}$ and proceed by induction on the number of points in $\{\alpha(p) > 0\}$. Reduce the problem for differentials to that for meromorphic functions by considering, in case $\mathfrak{B}_2(\alpha)$ has a member ω not the zero differential, the space of f satisfying

$$\partial f(p) + [\partial(p; \omega) + \alpha(p)] \geq 0, \quad p \in S,$$

and noting that the dimension of the space of such f is equal to the dimension of $\mathfrak{B}_2(\alpha)$. We are presupposing the usual standard elementary facts concerning linear vector spaces and linear transformations (cf. [58]).

Our object in introducing this exercise and Exs. 5.3 and 5.4 is to state (without proof) the Riemann–Roch theorem, which is one of the most important theorems concerning compact Riemann surfaces:

$$\dim \mathfrak{B}_1(-\alpha) = \dim \mathfrak{B}_2(\alpha) - \sum_{p \in S} \alpha(p) - \frac{\chi(S)}{2}. \quad (5.3)$$

Here “dim” means “dimension.” For proofs, cf. [2], [113].

It is interesting to see what information is yielded by the Riemann–Roch theorem. On taking α as the constant zero, we see that the left side of (5.3) is 1 and thus the dimension of the space of abelian differentials on S with divisors taking nonnegative values (the analytic abelian differentials) is $[2 + \chi(S)]/2$.

If $\alpha(a) \leq -[1 + \chi(S)]$, $\alpha(p) = 0$ for $p \neq a$, then $\dim \mathfrak{B}_2(\alpha) = 0$ and

$$\dim \mathfrak{B}_1(-\alpha) = -[2\alpha(a) + \chi(S)]/2 \geq [2 + \chi(S)]/2.$$

In particular, when $\alpha(a) = -[1 + \chi(S)]$, $\alpha(p) = 0$ for $p \neq a$, then

$$\dim \mathfrak{B}_1(-\alpha) = [2 + \chi(S)]/2.$$

By considering the linear transformation introduced above taking f into $(c_{-k}(f))_{k \in \mathbf{N}}$, which is $(1, 1)$ on $\mathfrak{B}_1(-\alpha)$ for this special choice of α , show that

$$\{-\partial_f(a): f \in \mathfrak{B}_1(-\alpha) - \{0\}\}$$

consists of precisely $[2 + \chi(S)]/2$ members of the section

$$\langle 0, 1 + \chi(S) \rangle$$

(Weierstrass gap theorem).

5.6. Show that in the case of **A** of Ex. 5.3 of this set there exists an abelian differential ω on **A** which has divisor everywhere zero. In fact, $(1/V) dC$ is such a differential.

5.7. Let $n \in \mathbf{N}$; let $b_1, \dots, b_{2(n+1)}$ be distinct complex numbers; let

$$a_0(z) = \prod_{k=1}^{2(n+1)} (z - b_k);$$

let P be the polynomial with zeroth component a_0 , second component the constant 1, and all others the constant 0. Show that P is an irreducible member of the ring of polynomials over $\mathbf{M}(\hat{\mathbf{C}})$. Let **A** be the analytic entity such that $(\mathbf{A}, C, V)$ annihilates P . Calculate $\chi(\mathbf{A})$ with the aid of dC . Verify the Riemann–Roch theorem for **A** in the special case where α is the constant zero and show that here the differentials $(C^k/V) dC$, $k = 0, \dots, n-1$, constitute a basis for $\mathfrak{B}_2(\alpha)$, the obvious convention holding when $n = 0$.

6. EXISTENCE THEOREM FOR AUXILIARY HARMONIC FUNCTIONS

Our object in this section is to establish the existence of a harmonic function of the kind specified in statement (5) of the preceding section. We approach

the question by first showing that subharmonicity may be appropriately introduced in the setting of Riemann surfaces and that the Perron method remains available, and thereupon treating the existence problem by Perron methods.

Given an open subset O of a Riemann surface S , we say that $u: D \rightarrow \mathbf{R}$, $O \subset D \subset S$, is *subharmonic in O* provided that u is continuous at each point of O and that for each uniformizer σ with domain a neighborhood of 0 and $\sigma(0) \in O$ the inequality

$$u[\sigma(0)] \leq \frac{1}{2\pi} \int_0^{2\pi} u[\sigma(re^{i\theta})] d\theta \quad (6.1)$$

holds for small positive r .†

It is immediate that many results of the plane theory extend. Thus harmonic functions, finite sums of subharmonic functions, the maximum of a finite set of subharmonic functions, and the product of a nonnegative number and a subharmonic function are all subharmonic. The maximum principle remains valid. So does the principle of the harmonic majorant when we consider properly contained *relatively compact* domains.

We associate with a uniformizer σ as above and r small and positive the (σ, r) *Poisson modification*, v , of u which is defined by

$$v(p) = u(p), \quad p \in D - \sigma[\Delta(0; r)],$$

$$v[\sigma(z)] = \frac{1}{2\pi} \int_0^{2\pi} u[\sigma(re^{i\theta})] k(\theta, z/r) d\theta, \quad |z| < r.$$

Clearly v is subharmonic in O . Given a family Φ of functions subharmonic on a region O (that is, having domain O and subharmonic in O), we term Φ a *Perron family* provided that (1) $\max\{u, v\} \in \Phi$ when $u, v \in \Phi$, (2) for each uniformizer σ , $\sigma(0) \in O$, for small positive r the (σ, r) Poisson modification of u belongs to Φ for each $u \in \Phi$. The basic theorem concerning Φ and its proof persist.

† This definition is readily seen to be concordant with that used in the case of the plane and finite-valued functions. It suffices to invoke the solution of the Dirichlet problem for a disk and the principle of the harmonic majorant. To show concordance we need only show that a function subharmonic in a plane open set O in the sense of Ch. XIII is subharmonic in O in the present sense, $\mathbf{C}$ being considered a Riemann surface with Θ consisting of the identity map of $\mathbf{C}$ onto itself. Here if $\sigma[\Delta(0; r)] \subset O$ and v is a continuous map of $\Delta(0; r)$ into $\mathbf{R}$, harmonic in $\Delta(0; r)$ and satisfying $v(z) = u[\sigma(z)]$, $|z| = r$, by the principle of the harmonic majorant $v[\sigma^{-1}(z)] \geq u(z)$, $z \in \sigma[\Delta(0; r)]$, and so $v(0) \geq u[\sigma(0)]$. The inequality (6.1) follows. On purely pragmatic grounds we are considering only finite-valued subharmonic functions. Of course, the theory may be considered under more general hypotheses.

EXERCISE

6.1. Treat the subharmonicity of a (σ, r) modification for the theory allowing functions taking the value $-\infty$.

The existence of auxiliary harmonic functions. Let S be a Riemann surface and let σ be a uniformizer for S with domain $\Delta(0; 2)$. We show

Theorem 6.1: *There exists u harmonic on $S - \{\sigma(0)\}$ such that*

$$u[\sigma(z)] = c \log |z| + \operatorname{Re} f(z),$$

$0 < |z| < 2$, where $c \in \mathbf{R}$ and f is analytic on $\{0 < |z| < 2\}$ and has a pole of order 1 at 0.

PROOF: We distinguish two cases.

Case 1: There exists for some r , $0 < r < 2$, a bounded function ω continuous on $S - \sigma[\Delta(0; r)]$ which takes the value 0 at each point of $\sigma[C(0; r)]$ and positive values at the points of $\Omega_r = S - \sigma[\overline{\Delta(0; r)}]$ and, in addition, is harmonic in Ω_r . We define for $c \in \mathbf{R}$ the function u_c with domain $S - \{\sigma(0)\}$ by

$$u_c[\sigma(z)] = \operatorname{Re} \left(\frac{r}{z} - \frac{z}{r} \right), \quad 0 < |z| \leq r,$$

$$u_c(p) = c \omega(p), \quad p \in \Omega_r.$$

We assert that u_c and $-u_{(-c)}$ are subharmonic on their domain for c large and positive. By the boundary maximum principle for harmonic functions

$$c\omega[\sigma(z)] \geq \left| \operatorname{Re} \left(\frac{r}{z} - \frac{z}{r} \right) \right|,$$

$r \leq |z| \leq (2+r)/2$, when c is large and positive. It suffices to test the mean-value property of u_c and $-u_{(-c)}$ at the points of $\sigma[C(0; r)]$ which is concluded on noting the harmonicity of $\operatorname{Re}(rz^{-1} - zr^{-1})$ and the fact that

$$u_{(-c)}[\sigma(z)] \leq \operatorname{Re} \left(\frac{r}{z} - \frac{z}{r} \right) \leq u_c[\sigma(z)],$$

$r \leq |z| \leq (2+r)/2$, c large and positive. We fix c (necessarily positive) so that u_c and $-u_{(-c)}$ are subharmonic on $S - \{\sigma(0)\}$ and let d be a number exceeding $2 \sup\{c \omega(p) : p \in \Omega_r\}$. We take Φ as the family of functions v subharmonic on $S - \{\sigma(0)\}$ and satisfying

$$v(p) - [u_{(-c)}(p) + d] \leq 0,$$

$p \in S - \{\sigma(0)\}$. We note that Φ is a Perron family and that $u_c \in \Phi$. With M denoting the upper envelope of $\Phi^\dagger$ we see that M is harmonic on $S - \{\sigma(0)\}$ and satisfies

$$u_c(p) \leq M(p) \leq u_{(-c)}(p) + d,$$

$p \in S - \{\sigma(0)\}$. That M has the desired local behavior near $\sigma(0)$ follows from the fact that a function h harmonic on $\Delta(0; \rho) - \{0\}$ and bounded is the restriction of the real part of a function analytic at 0 as we see on applying (3.3) of Ch. XI and estimating the complex Fourier coefficients of $h(se^{i\theta})$, $0 < s < \rho$. We even are assured of the existence of a function u for which the coefficient of $\log |z|$ is 0.

Case 2: The assumption of Case 1 does not hold. We first establish under the Case 2 hypothesis a maximum principle for $\bar{\Omega}_r$, $0 < r < 2$, namely:

If u is a bounded continuous map of $\bar{\Omega}_r$ into $\mathbf{R}$, harmonic in Ω_r , then u attains its maximum value on $\sigma[C(0; r)]$.

To see this we proceed as follows. Let A denote the infimum and B the supremum of the image of u . We define v_1 by requiring that $v_1[\sigma(z)]$ be equal to the value at z of the solution h_1 of the Dirichlet problem for the annulus $\{r < |z| < (2+r)/2\}$ satisfying both $h_1(z) = u[\sigma(z)]$, $|z| = r$, and $h_1(z) = A$, $|z| = (2+r)/2$, and that $v_1(p) = A$, $p \in \Omega_{(2+r)/2}$. We define v_2 analogously with B replacing A . Clearly v_1 and $-v_2$ are subharmonic in Ω_r . Now let Φ_1 denote the family of functions w subharmonic on Ω_r which satisfy

$$w(p) \leq \min\{u(p), \mu\},$$

$p \in \Omega_r$, where $\mu = \max\{u[\sigma(z)]: |z| = r\}$. Let Φ_2 denote the family of functions w subharmonic on Ω_r which satisfy

$$w(p) \leq v_2(p),$$

$p \in \Omega_r$. Both Φ_1 and Φ_2 are Perron families. We let M_k denote the upper envelope of Φ_k , $k = 1, 2$. Since $v_1|_{\Omega_r} \in \Phi_1$ and $u|_{\Omega_r} \in \Phi_2$, we conclude that

$$v_1(p) \leq M_1(p) \leq u(p) \leq M_2(p) \leq v_2(p),$$

$p \in \Omega_r$. Since $M_2 - M_1$ is a bounded function, harmonic on Ω_r , taking non-negative values, and having the limit zero at each point of $\sigma[C(0; r)]$, $M_2 = M_1$. Hence $u(p) = M_1(p) \leq \mu$, $p \in \Omega_r$. The maximum principle in question is thereby established. Obviously a corresponding minimum principle holds.

Actually the harvest from this argument is greater than it appears. Indeed, without reference to the dichotomy between Case 1 and Case 2, we see that if f is a continuous map of $\sigma[C(0; r)]$ into $\mathbf{R}$, if A is the minimum and B the maximum of the image of f , if h_k , $k = 1, 2$, are defined as above with f replacing u and the present A and B those used above, and if thereupon v_1, v_2 are

† That is, $M(p) = \sup\{v(p): v \in \Phi\}$.

introduced in terms of the present h_1, h_2, A, B , we see that Φ_2 taken relative to the present v_2 is a Perron family containing $v_1|_{\Omega_r}$, so that its upper envelope M satisfies $v_1(p) \leq M(p) \leq v_2(p)$, $p \in \Omega_r$. We see that $f \cup M$ is bounded, continuous on $\bar{\Omega}_r$, and is harmonic in Ω_r . Under the Case 2 assumption we see by considering the difference of candidate functions that there is exactly one bounded continuous map of $\bar{\Omega}_r$ into $\mathbf{R}$, harmonic in Ω_r , whose restriction to $\sigma[C(0; r)]$ is f .

We continue by constructing a function u of the theorem, assuming the Case 2 hypothesis. To that end let $0 < r < 1$ and let U_r denote the unique bounded continuous map of $\bar{\Omega}_r$ into $\mathbf{R}$, harmonic in Ω_r , and satisfying $U_r[\sigma(z)] = \operatorname{Re}(1/z)$, $|z| = r$. We see that the minimum value taken by U_r on $\sigma[C(0; 1)]$ is actually less than the maximum value taken by U_r on $\sigma[C(0; 1)]$. For otherwise, because of the Case 2 hypothesis, U_r would be a constant. We introduce

$$V_r(p) = \alpha(r) U_r(p) + \beta(r),$$

with domain $\bar{\Omega}_r$, $\alpha(r)$, $\beta(r)$ being real numbers such that the maximum value taken on $\sigma[C(0; 1)]$ by V_r is 1 and the minimum value taken on $\sigma[C(0; 1)]$ by V_r is 0. The family of functions

$$v_r = V_r|_{\sigma(\{1 < |z| < 2\})},$$

$0 < r < 1$, is normal. There exists a decreasing sequence $(r(n))_{n \in \mathbf{N}}$ where $0 < r(0) < 1$, $\lim_{n \rightarrow \infty} r(n) = 0$, such that $(v_{r(n)})$ is uniformly Cauchy in $\sigma(\{1 < |z| < 2\})$. By virtue of the Case 2 hypothesis

$$(V_{r(n)}|_{\Omega_1})$$

is uniformly Cauchy in Ω_1 . We denote the function to which this sequence tends by V .

Now we introduce the representation of $V_r[\sigma(z)]$, $r < |z| < 2$, of the form

$$c(r) \log |z| + \operatorname{Re} \sum_{-\infty}^{+\infty} a_k(r) z^k,$$

where $a_0(r)$ is real, assured by (3.3), Ch. XI, and deduce the following formulas involving the coefficients:

$$c(r) \log |s| + a_0(r) = \frac{1}{2\pi} \int_0^{2\pi} V_r[\sigma(se^{i\theta})] d\theta, \quad (6.2)$$

$r \leq s < 2$, and

$$a_k(r) s^k + \overline{a_{-k}(r)} s^{-k} = \frac{1}{\pi} \int_0^{2\pi} V_r[\sigma(se^{i\theta})] e^{-ki\theta} d\theta, \quad (6.3)$$

$r \leq s < 2$, k a positive integer. From (6.3) we obtain for $k \geq 2$, $s = r$, the equalities

$$a_k(r) r^{2k} + \overline{a_{-k}(r)} = 0,$$

and for $k \geq 2$, $s = 1$, the inequalities

$$|a_k(r) + \overline{a_{-k}(r)}| \leq 2,$$

whence we deduce that

$$|a_k(r)| \leq \frac{2}{1-r} \quad (6.4)$$

and

$$|a_{-k}(r)| \leq \frac{2r^{2k}}{1-r}, \quad (6.5)$$

$k \geq 2$.

We next consider the corresponding representation of $V[\sigma(z)]$, $1 < |z| < 2$, namely

$$c \log |z| + \operatorname{Re} \sum_{-\infty}^{+\infty} a_k z^k, \quad (6.6)$$

where a_0 is real. We note that

$$a_k = \lim_{n \rightarrow \infty} a_k[r(n)], \quad k \in \mathbb{Z},$$

and

$$c = \lim_{n \rightarrow \infty} c[r(n)],$$

as we see by appeal to (6.2) and (6.3) considering two distinct values of s in $]1, 2[$. Hence by (6.5) we have $a_{-k} = 0$, $k \geq 2$. It follows that

$$V[\sigma(z)] = c \log |z| + \operatorname{Re}(a_{-1}/z) + \operatorname{Re} \sum_0^{\infty} a_k z^k, \quad (6.7)$$

and hence there exists u harmonic on $S - \{\sigma(0)\}$ such that $u(p) = V(p)$, $p \in \Omega_1$, and $u[\sigma(z)]$ is equal to the right side of (6.7) for $0 < |z| \leq 1$. Now $a_{-1} \neq 0$, for otherwise u would possess a limit, finite or infinite, at $\sigma(0)$. By the maximum principle holding under the Case 2 hypothesis we would conclude that u is constant. But there exists a point of $\sigma[C(0; 1)]$ at which u attains the value 0 and a second point of $\sigma[C(0; 1)]$ at which u attains the value 1, for $(V_{r(n)})$ tends uniformly to u on $\sigma[C(0; 1)]$ as we see by introducing ρ , $r(0) < \rho < 1$,

and estimating the size of

$$|V_r[\sigma(z)] - u[\sigma(z)]|, \quad (6.8)$$

$|z| = \rho$, $0 < r < \rho$, with the aid of (6.4) and (6.5). In fact, (6.8) does not exceed

$$|c(r) - c| \log \frac{1}{\rho} + \sum_{k=1}^m |a_k(r) - a_k| \rho^k + \frac{4}{1-r} \frac{\rho^{m+1}}{1-\rho} + 2 \left(\frac{r}{1-r} \right)^2, \\ m = 1, 2, \dots$$

On setting $r = r(n)$ here we obtain a sequence with limit $4\rho^{m+1}/(1-\rho)$. Given the arbitrariness of m , we see that $(V_{r(n)})$ does tend uniformly to u on $\sigma[C(0; 1)]$. Theorem 6.1 follows.

7. FIELD ISOMORPHISM AND CONFORMAL EQUIVALENCE OF COMPACT RIEMANN SURFACES

Questions relating algebraic isomorphism of the rings of analytic functions and the conformal equivalence of their domains were studied earlier. We recall the theorem of Bers treated in Ch. VIII. At that point the recent remarkable and important results of Hej Iss'sa concerning noncompact Riemann surfaces whose fields of meromorphic functions are isomorphic were noted. In the present section we shall establish a classical isomorphism theorem for compact Riemann surfaces and shall also show that at the least field structure distinguishes compact Riemann surfaces from noncompact Riemann surfaces.

We turn at once to the isomorphism theorem for compact Riemann surfaces.

Theorem 7.1: *Let S and T be compact Riemann surfaces and let α be an isomorphism of $\mathbf{M}(T)$, the field of functions meromorphic on T , onto $\mathbf{M}(S)$, the field of functions meromorphic on S . Suppose that α preserves the complex constants (cf. §2, Ch. VIII). Then there exists a unique map σ of S into T such that*

$$\alpha(f) = f \circ \sigma, \quad (7.1)$$

$f \in \mathbf{M}(T)$. Further the map σ is a univalent analytic map of S onto T .

PROOF: The *uniqueness* aspect of the question is immediate since given $a, b (\neq a) \in T$, there exists $f \in \mathbf{M}(T)$ such that $f(a) \neq f(b)$. Contrapositively, if $f(a) = f(b)$ for all $f \in \mathbf{M}(T)$, then $a = b$. The uniqueness of σ follows.

Existence. Let $f, g \in \mathbf{M}(T)$, where f is not constant, and suppose that (T, f, g) is such that the component φ of $(\mathbf{A}, \varphi)$ associated with it by Lemma 2.2, this chapter, maps T univalently onto $\mathbf{A}$. Such a choice of f and g is possible by the existence of nonconstant members of $\mathbf{M}(T)$ and the affirmative solution of the problem of Riemann and Klein. Let n denote the common value taken by the valence of f . We have

$$g^n + \sum_0^{n-1} (a_k \circ f)g^k = 0$$

where $P = (a_k)$ is the unique member of $\mathfrak{Q}$ (cf. Theorem 3.2, this chapter) annihilated by $(\mathbf{A}, C, V)$. Hence since α is an isomorphism preserving the complex constants, we see that $(S, \alpha(f), \alpha(g))$ annihilates P . Since $\mathbf{A}$ is the unique analytic entity which taken with its center and value function annihilates P , we see that $(S, \alpha(f), \alpha(g))$ is associated with $(\mathbf{A}, \varphi_1)$ by Lemma 2.2, this chapter, where now φ_1 is an analytic map of S into $\mathbf{A}$. Setting $\sigma = \varphi^{-1} \circ \varphi_1$, we obtain by the mediation of $\alpha(f) = C \circ \varphi_1$ and $f = C \circ \varphi$ the equality $\alpha(f) = f \circ \sigma$ and similarly we find $\alpha(g) = g \circ \sigma$. If $h \in \mathbf{M}(T)$, we have

$$h = \sum_0^{n-1} (b_k \circ f)g^k$$

where $b_k \in \mathbf{M}(\hat{\mathbf{C}})$, $k = 0, \dots, n-1$. Hence

$$\alpha(h) = \sum_0^{n-1} \{b_k \circ [\alpha(f)]\}[\alpha(g)]^k = h \circ \sigma.$$

A corresponding result holds for α^{-1} yielding an analytic map τ of T into S . By the uniqueness result we see that $\sigma \circ \tau$ is the identity map of T onto itself since $f \circ (\sigma \circ \tau) = f$, $f \in \mathbf{M}(T)$, and similarly $\tau \circ \sigma$ is the identity map of S onto itself. We conclude that σ is a univalent map of S onto T . The theorem follows.

REMARK: It is to be noted that an arbitrary univalent analytic map σ of a Riemann surface S onto a Riemann surface T yields the isomorphism of $\mathbf{M}(T)$ onto $\mathbf{M}(S)$ which assigns $f \circ \sigma$ to $f \in \mathbf{M}(T)$.

EXERCISE

7.1. Show that if (S, Θ) is a Riemann surface and $\tilde{\Theta}$ denotes the set of maps $\{(\tilde{z}, \theta(z)): z \in \text{domain of } \theta\}$, $\theta \in \Theta$, then $(S, \tilde{\Theta})$ is a Riemann surface.

Apply this result to show that if α is an isomorphic map of $\mathbf{M}(T)$ onto $\mathbf{M}(S)$, where S and T are compact, which maps the complex constants onto their conjugates,

then $\alpha(f) = f \circ \sigma$, $f \in \mathbf{M}(T)$, where f is the composition of the conjugate map of $\hat{\mathbf{C}}$ onto itself with f and σ is now a univalent map of S onto T , analytic in the sense of the conformal structure $\bar{\theta}$ on S .

On the behavior of constants under isomorphisms. If we consider an arbitrary isomorphism α of $\mathbf{M}(T)$ into $\mathbf{M}(S)$, where now we drop the compactness conditions on S and T , we see on referring to the proof of the corresponding fact in the proof of the theorem of Bers that α maps the constants of $\mathbf{M}(T)$ into the constants of $\mathbf{M}(S)$. When α maps $\mathbf{M}(T)$ onto $\mathbf{M}(S)$, then α maps the set of constants of $\mathbf{M}(T)$ onto the set of constants of $\mathbf{M}(S)$.

This is an appropriate moment to recall that there exist highly complicated isomorphisms of $\mathbf{C}$ onto itself [139]. Such isomorphisms may be constructed with the aid of transcendental field extensions. In the theorem of Bers the isomorphism of $\mathbf{C}$ onto itself which emerges from the given isomorphism of rings of analytic functions turns out to be either the identity map or the conjugate map.

We shall now give an example of compact Riemann surfaces whose associated fields of meromorphic functions admit an isomorphism of one onto the other preserving the constant i but which are not conformally equivalent. Suppose that α is a "bad" isomorphism of $\mathbf{C}$ onto itself. We may, of course, assume that $\alpha(i) = i$. Let $c \in \mathbf{R}$ be such that $\text{Im } \alpha(c) \neq 0$. Let $P = (a_k)$ where $a_0(z) = z(z-1)(z-c)$; $a_2(z) = -1$; $a_k = 0$ for $k \neq 0, 2$. Let $Q = (b_k)$ where $b_0(z) = z(z-1)[z-\alpha(c)]$, $b_2(z) = -1$, $b_k = 0$, $k \neq 0, 2$. Let $(\mathbf{A}, C, V)$ annihilate P and $(\mathbf{B}, C_1, V_1)$ annihilate Q . Let β denote the unique isomorphism of $\mathbf{M}(\hat{\mathbf{C}})$ onto itself taking the constant $a_{\hat{\mathbf{C}}}$ into the constant $\alpha(a_{\hat{\mathbf{C}}})$ and keeping fixed the identity map of $\hat{\mathbf{C}}$ onto itself. The set of ordered pairs

$$(r_0 \circ C + (r_1 \circ C)V, \beta(r_0) \circ C_1 + [\beta(r_1) \circ C_1]V_1),$$

$r_0, r_1 \in \mathbf{M}(\hat{\mathbf{C}})$, is an isomorphism of $\mathbf{M}(\mathbf{A})$ onto $\mathbf{M}(\mathbf{B})$.

To avoid difficulties that are inessential for our present discussion we shall take advantage of the connection between the analytic entities considered and appropriate Weierstrass p -functions. Thus we let $c = \lambda(\omega)$, where λ is the modular function of §1, Ch. XVI, and let p denote the Weierstrass p -function associated with the additive subgroup of $\mathbf{C}$ generated by $\{1, \omega\}$. Now $\mathbf{A}$ is associated with $(\mathbf{C}, f, df')$ in the sense of Lemma 2.2 of the present chapter where

$$f(z) = \frac{p(z) - p(1/2)}{p(\omega/2) - p(1/2)},$$

and

$$d^2 = \frac{1}{4[p(\omega/2) - p(1/2)]}.$$

Let φ denote the cognate map of $\mathbf{C}$ onto $\mathbf{A}$. Suppose that ψ is a univalent analytic map of $\mathbf{A}$ onto $\mathbf{B}$. Then $g = C_1 \circ \psi \circ \varphi$ satisfies

$$g(z) = c_1 f(z + c_2) + c_3,$$

$z \in \mathbf{C}$, where the c_k are complex numbers, $c_1 \neq 0$. Now the linear map taking z into $c_1 z + c_3$ maps $\{0, 1, c\}$ onto $\{0, 1, \alpha(c)\}$. Hence $\alpha(c) \in \mathbf{R}$. The contradiction is apparent. We conclude that $\mathbf{A}$ is not conformally equivalent to $\mathbf{B}$.

We now turn to the second theorem, which states that field structure distinguishes compact Riemann surfaces from noncompact Riemann surfaces. We suppose, as we may, that the compact surface is an analytic entity with C not constant and show

Theorem 7.2: *Let $\mathbf{M}(S)$, where S is a given Riemann surface, be isomorphic to $\mathbf{M}(A)$, where A is a compact analytic entity. Then S is compact.*

PROOF: Let α be an isomorphism of $\mathbf{M}(S)$ onto $\mathbf{M}(A)$. Let $f = \alpha^{-1}(C)$, $g = \alpha^{-1}(V)$. From the compactness of $\mathbf{A}$, we see that

$$g^n + \sum_0^{n-1} (r_k \circ f) g^k = 0,$$

where $r_k \in \mathbf{M}(\hat{\mathbf{C}})$, $k = 0, \dots, n-1$, n being the constant value of the valence of C . Hence (S, f, g) induces $(\mathbf{B}, \varphi)$ in the sense of Lemma 2.2, this chapter, where $\mathbf{B}$ is compact. Given $h \in \mathbf{M}(S)$, we see that

$$h = \sum_0^{n-1} (\rho_k \circ f) g^k,$$

where $\rho_k \in \mathbf{M}(\hat{\mathbf{C}})$, $k = 0, \dots, n-1$. Hence the map assigning to $F \in \mathbf{M}(\mathbf{B})$ the function $F \circ \varphi$ is an isomorphism of $\mathbf{M}(\mathbf{B})$ onto $\mathbf{M}(S)$. Now φ is univalent, for when $\varphi(a) = \varphi(b)$, $h(a) = h(b)$ for all $h \in \mathbf{M}(S)$ and, since some member of $\mathbf{M}(S)$ distinguishes two given distinct points of S , we see that $a = b$.

Suppose that $\varphi(S) \neq \mathbf{B}$. Then $\text{fr } \varphi(S) \neq \emptyset$. Let $b \in \text{fr } \varphi(S)$ and let (a_n) be a univalent map of $\mathbf{N}$ into $\varphi(S)$ satisfying $\lim_{n \rightarrow \infty} a_n = b$. Via quotients of differentials, we construct f_n meromorphic on $\mathbf{B}$ with a pole at a_n and elsewhere analytic save possibly at points of a finite set independent of n . We generate the differentials in question from harmonic functions of the kind whose existence is assured by Theorem 6.1, this chapter. It is clear that the construction may be so controlled that the totality of values taken by the divisors of the f_n has a finite minimum. We assume that this is the case. By a suitable choice of complex c_n we are assured that

$$\sum_0^{\infty} c_n f_n |(\mathbf{B} - \{b\})$$

is meromorphic and has poles at each a_n , n large. Composing this function with φ we obtain a member of $\mathbf{M}(S)$ with infinitely many poles. This was, of course, the purpose of the construction. Since every member of $\mathbf{M}(S)$ is of the form $F \circ \varphi$, $F \in \mathbf{M}(\mathbf{B})$, it has bounded valence when it is not constant. Contradiction. Hence $\varphi(S) = \mathbf{B}$. In particular, S is compact.